

HCDE 511

Information Visualization

Winter 2026

7
Week

Week

Multivariate display of data

20 min Break

Multivariate cars activity

5 min Break

**Multivariate project brainstorm &
sketch**

Week 1

Intro

2

Data
encoding &
mapping

3

Working
with data

4

Tableau
data &
dashboard

Visual
perception

5

Tasks &
inter-
actions

6

Collabora-
tive
critique

7

Multi-
variate
analysis

8

Trees,
networks
Animation,
distortion

9

Animation,
distortion,
story-
telling

10

Final pre-
sentations

Finals

A1 Good/bad

A2 Static vis

A3 Exploratory analysis

Project proposal

Rev. prop.

Final presentation

Final project writeup

Multivariate Analysis

Multivariate

Multiple quantitative values

Analytic variables / targets of analysis

Not indices, scales, or constants, like time

Shown in the same view, not side by side or view switching

(Not necessarily dependent variables)

Model	Origin	Year	Cylinders	Horsepower	MPG	Weight
amc ambassador dpl	US	70	8	190	15	3,850
chevrolet vega (sw)	US	71	4	72	22	2,408
chevrolet vega	US	72	4	90	20	2,408
ford pinto runabout	US	72	4	86	21	2,226
mazda rx2 coupe	Japan	72	3	97	19	2,330
amc matador (sw)	US	72	8	150	15	3,892
ford pinto (sw)	US	72	4	86	22	2,395
dodge coronet custom	US	73	8	150	15	3,777
toyota carina	Japan	73	4	88	20	2,279
chevrolet vega	US	73	4	72	21	2,401
datsum 610	Japan	73	4	94	22	2,379
ford pinto	US	73	4	85	19	2,310
chevrolet chevelle malibu classic	US	74	6	100	16	3,781
buick century	US	75	6	110	17	3,907
chevroelt chevelle malibu	US	75	6	105	16	3,897
plymouth fury	US	75	6	95	18	3,785
amc matador	US	76	8	120	15.5	3,962
mercedes-benz 280s	Europe	76	6	120	16.5	3,820
chevrolet caprice classic	US	77	8	145	17.5	3,880
chevrolet caprice classic	US	79	8	130	17	3,840
mercury grand marquis	US	79	8	138	16.5	3,955
dodge st. regis	US	79	8	135	18.2	3,830
vw rabbit	Europe	80	4	76	41.5	2,144
mazda glc	Japan	80	4	65	46.6	2,110
vw pickup	Europe	82	4	52	44	2,130

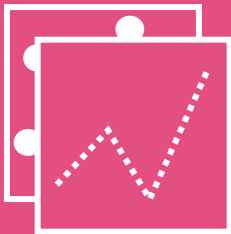
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vw rabbit	Europe	80	4	76	41.5	2,144
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vw pickup	Europe	82	4	52	44	2,130

Which items are most alike?

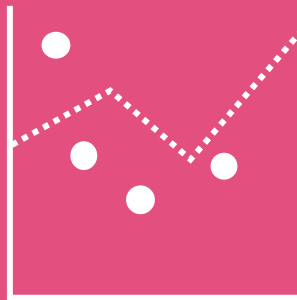
Which items are most exceptional?

How can items be combined into logical groups based on similarity (clustering)?

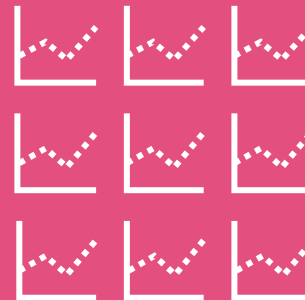
Which multivariate data profile corresponds best to a particular outcome (question)?



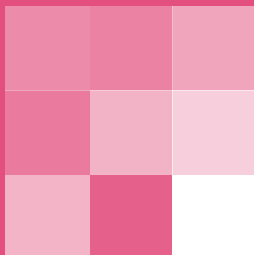
View Switching



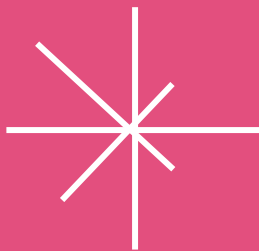
Layering



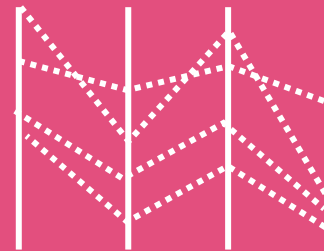
Small Multiples



Heatmaps



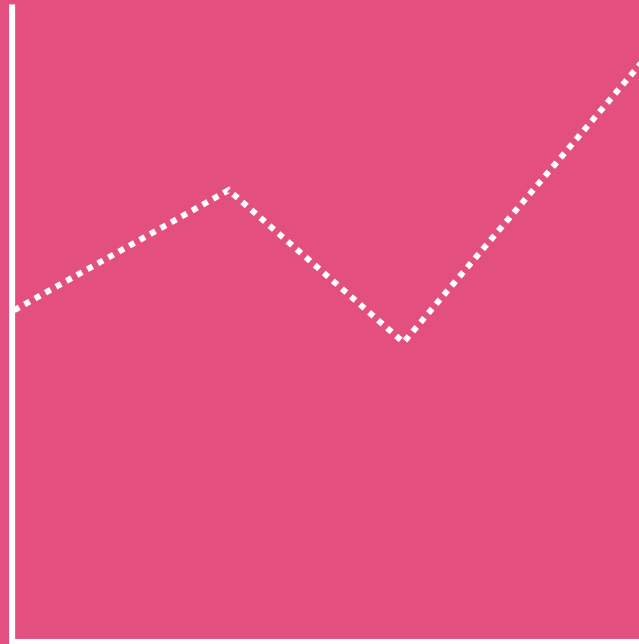
Glyphs



Parallel Coordinates

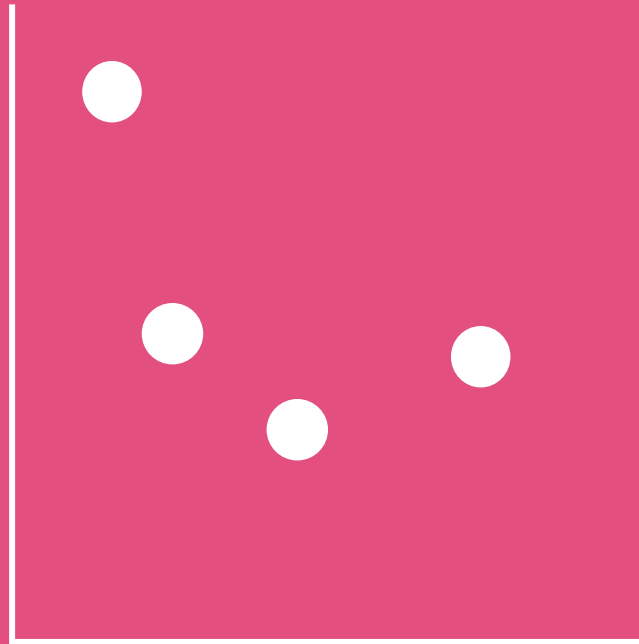
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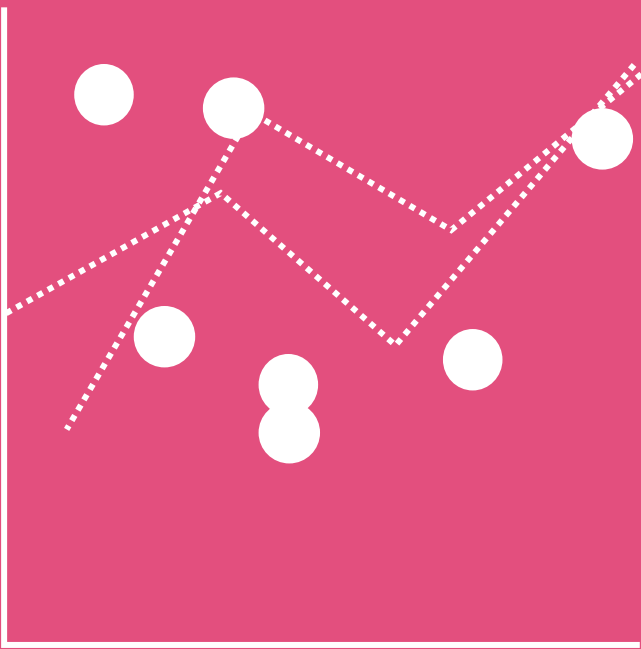
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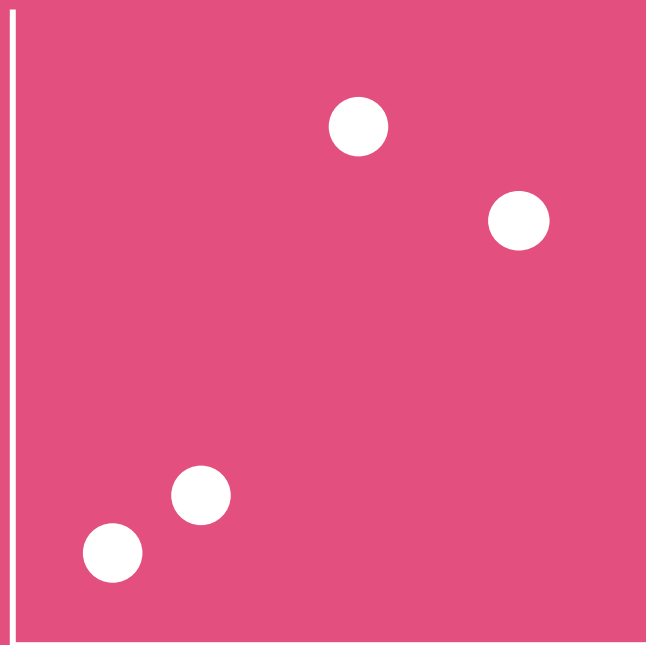
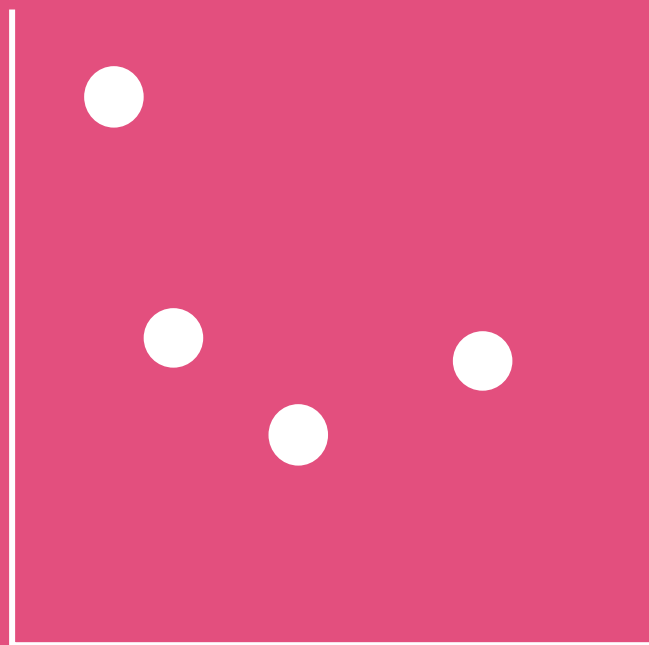
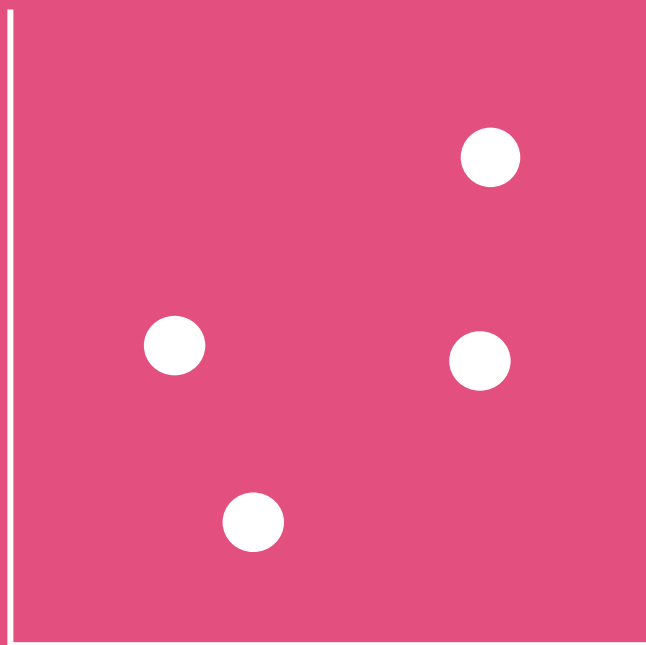


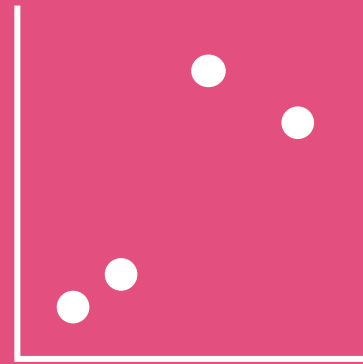
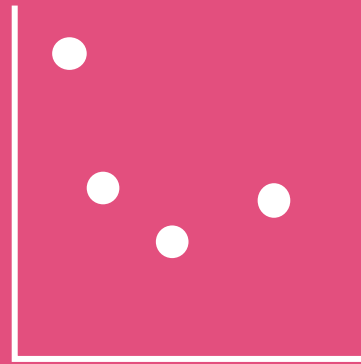
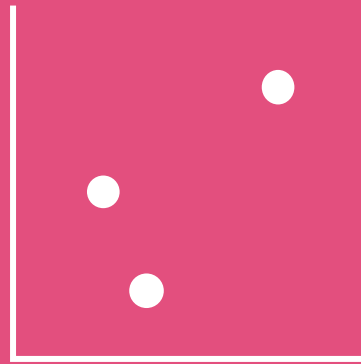
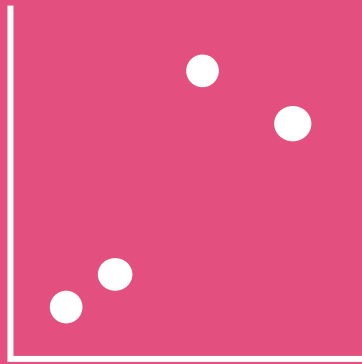
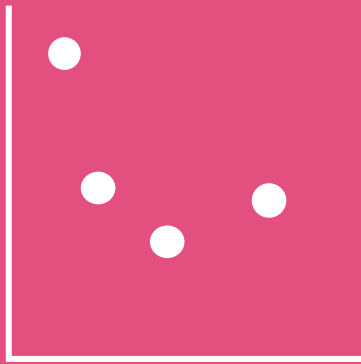
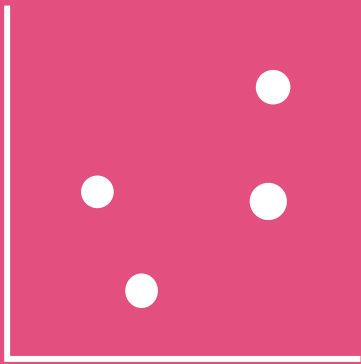
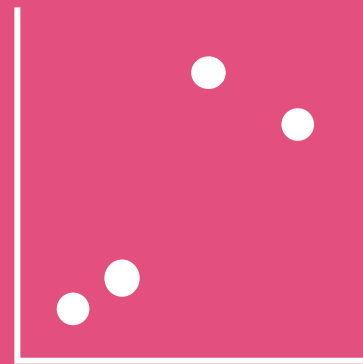
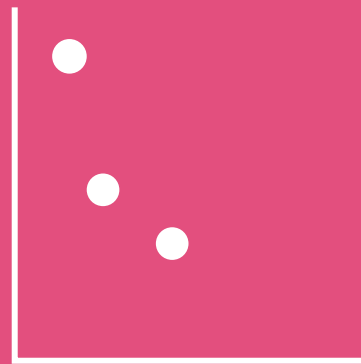
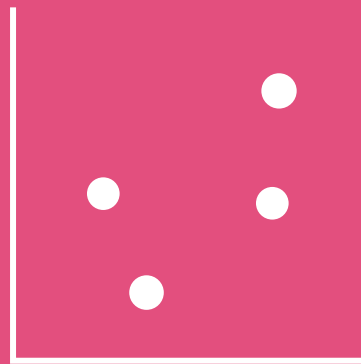
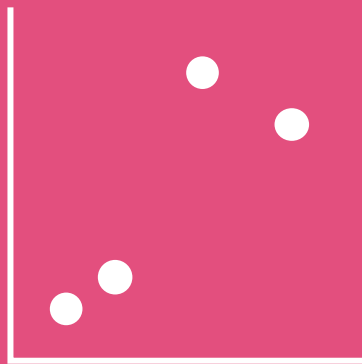
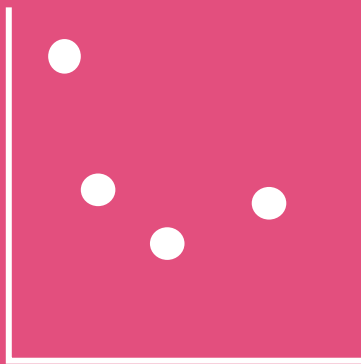
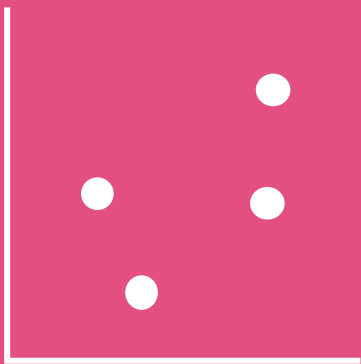
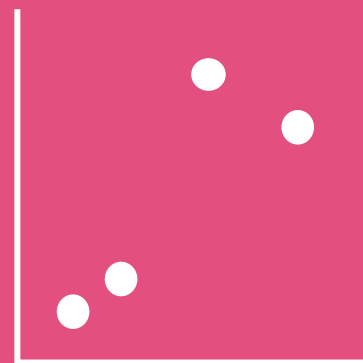
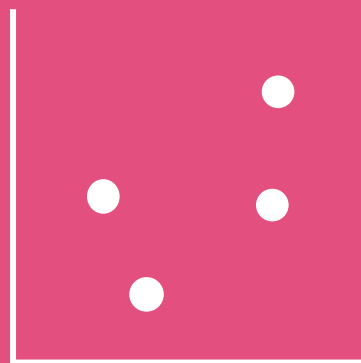
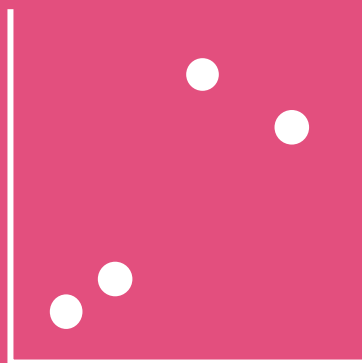
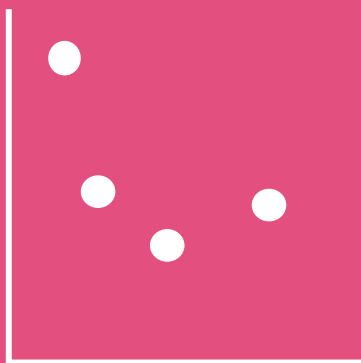
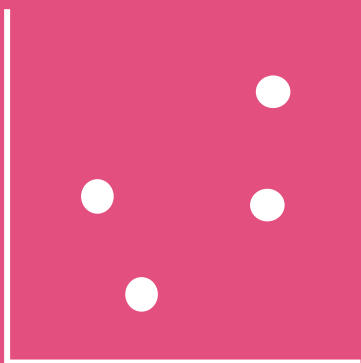
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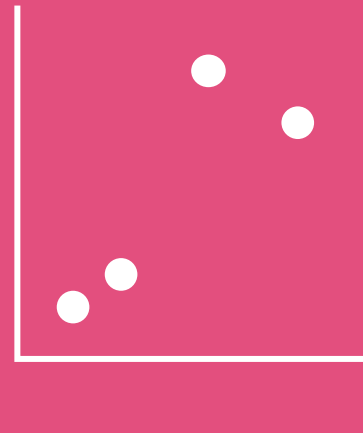
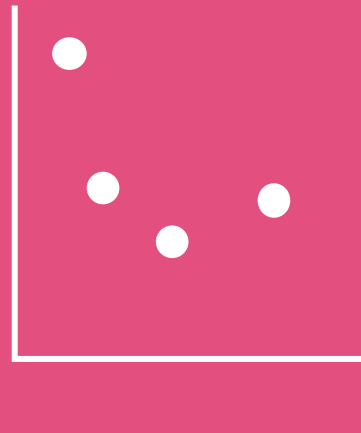
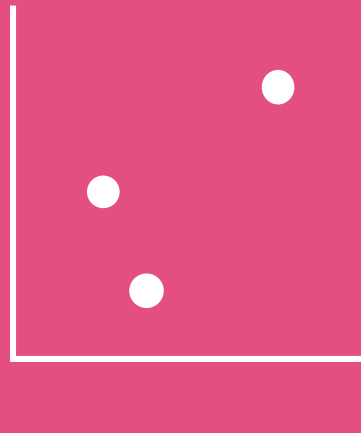
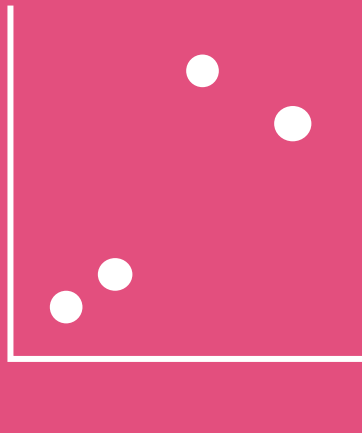
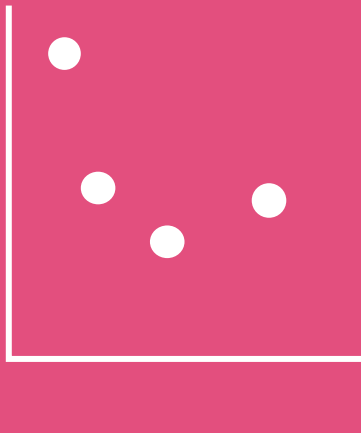
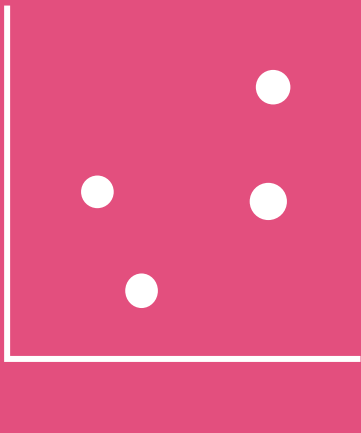
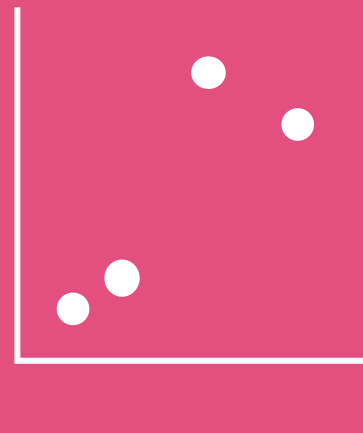
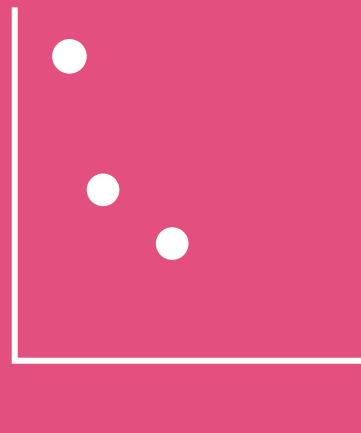
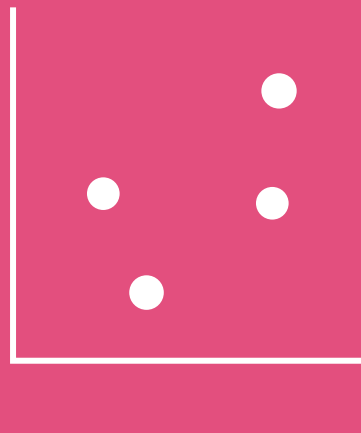
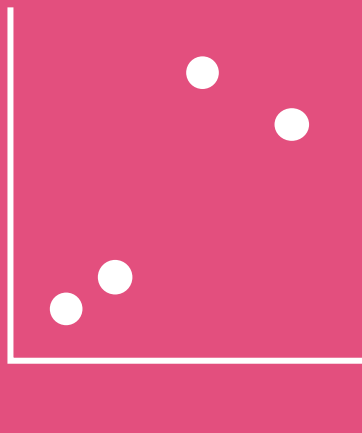
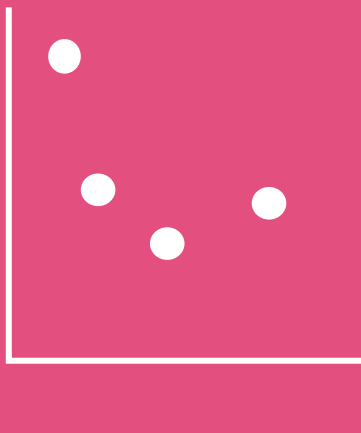
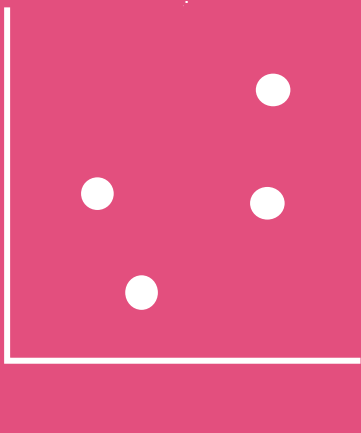
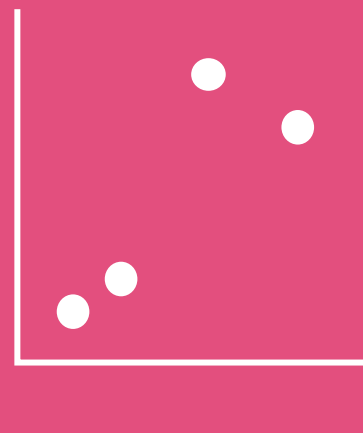
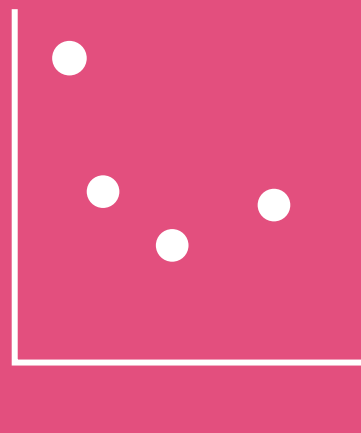
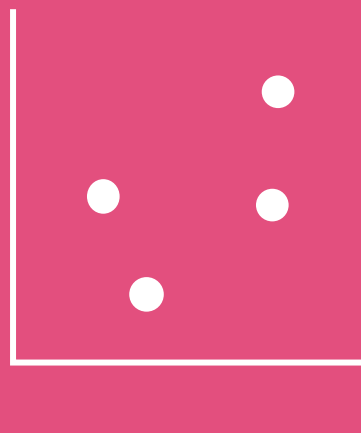
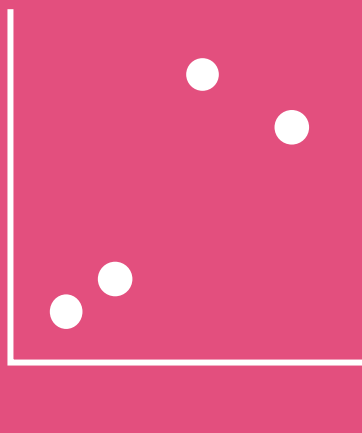
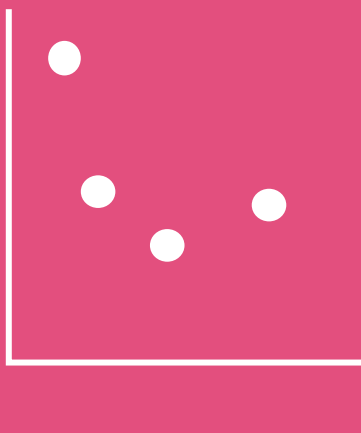
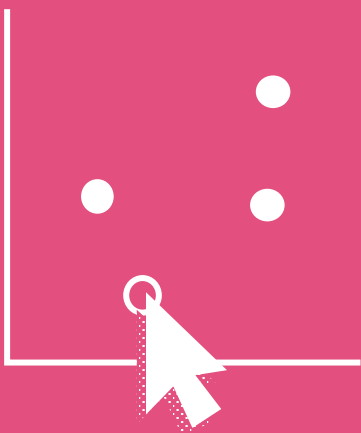
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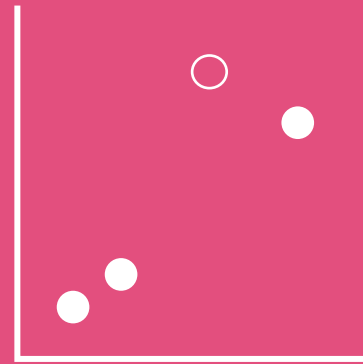
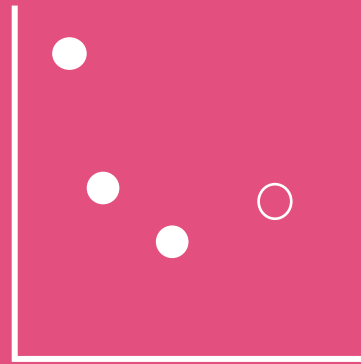
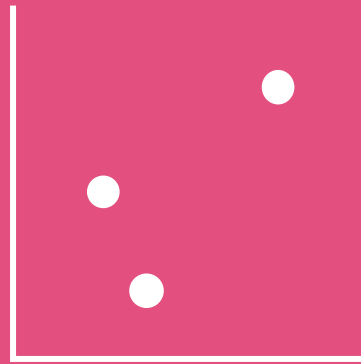
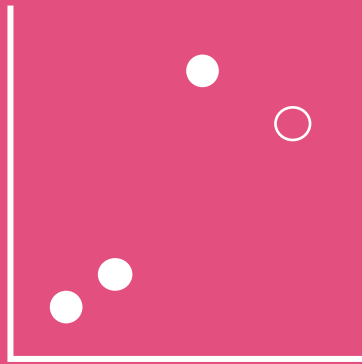
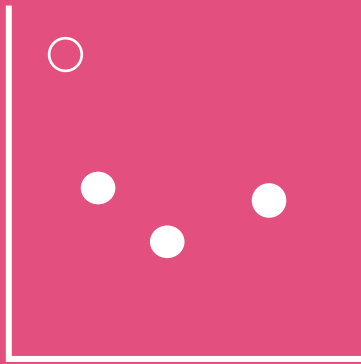
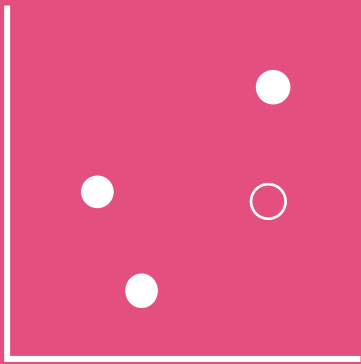
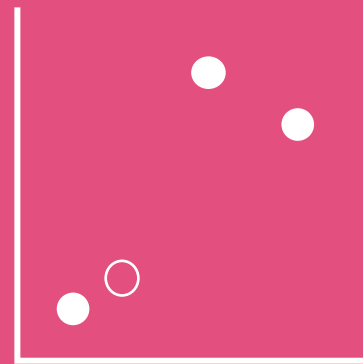
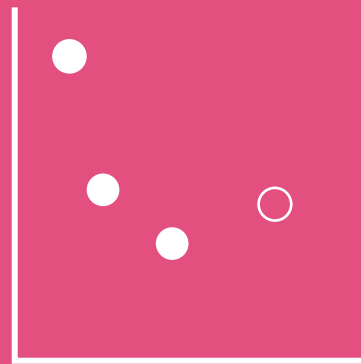
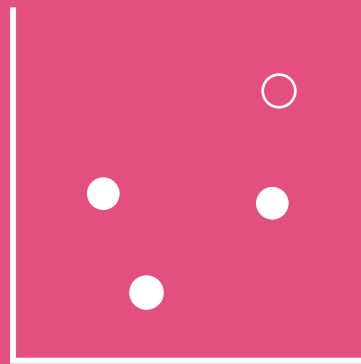
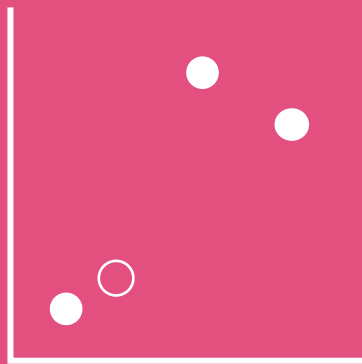
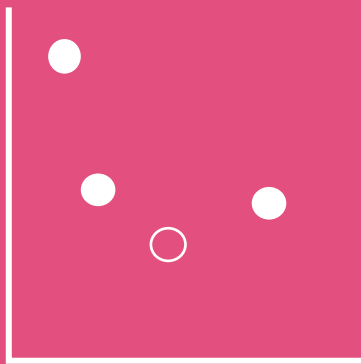
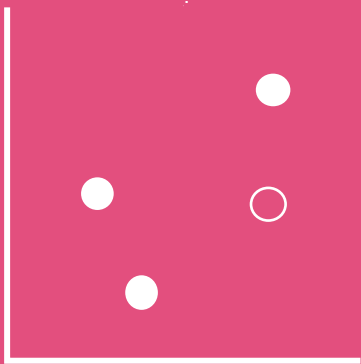
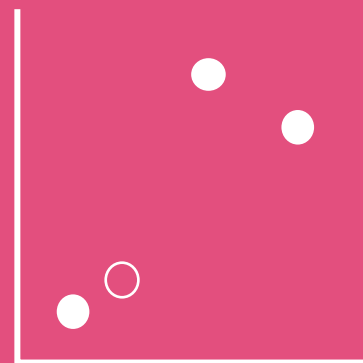
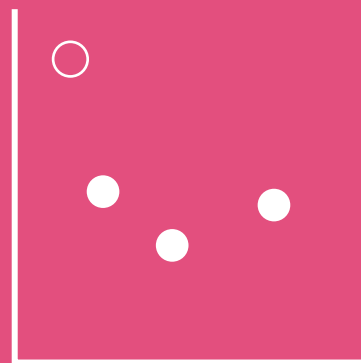
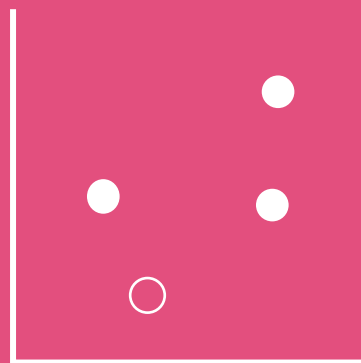
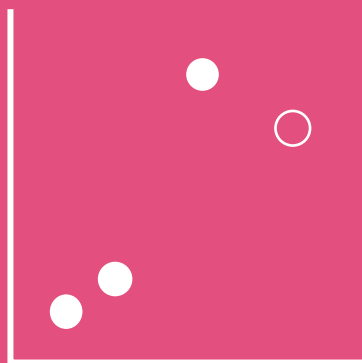
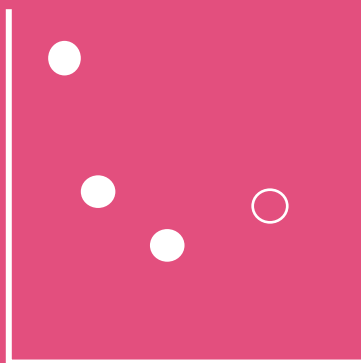
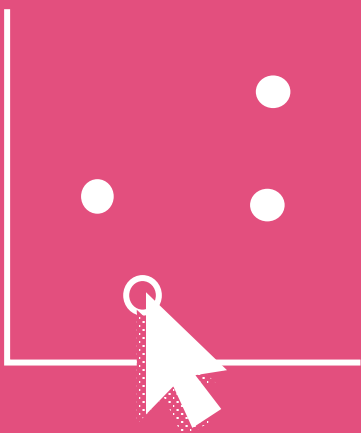




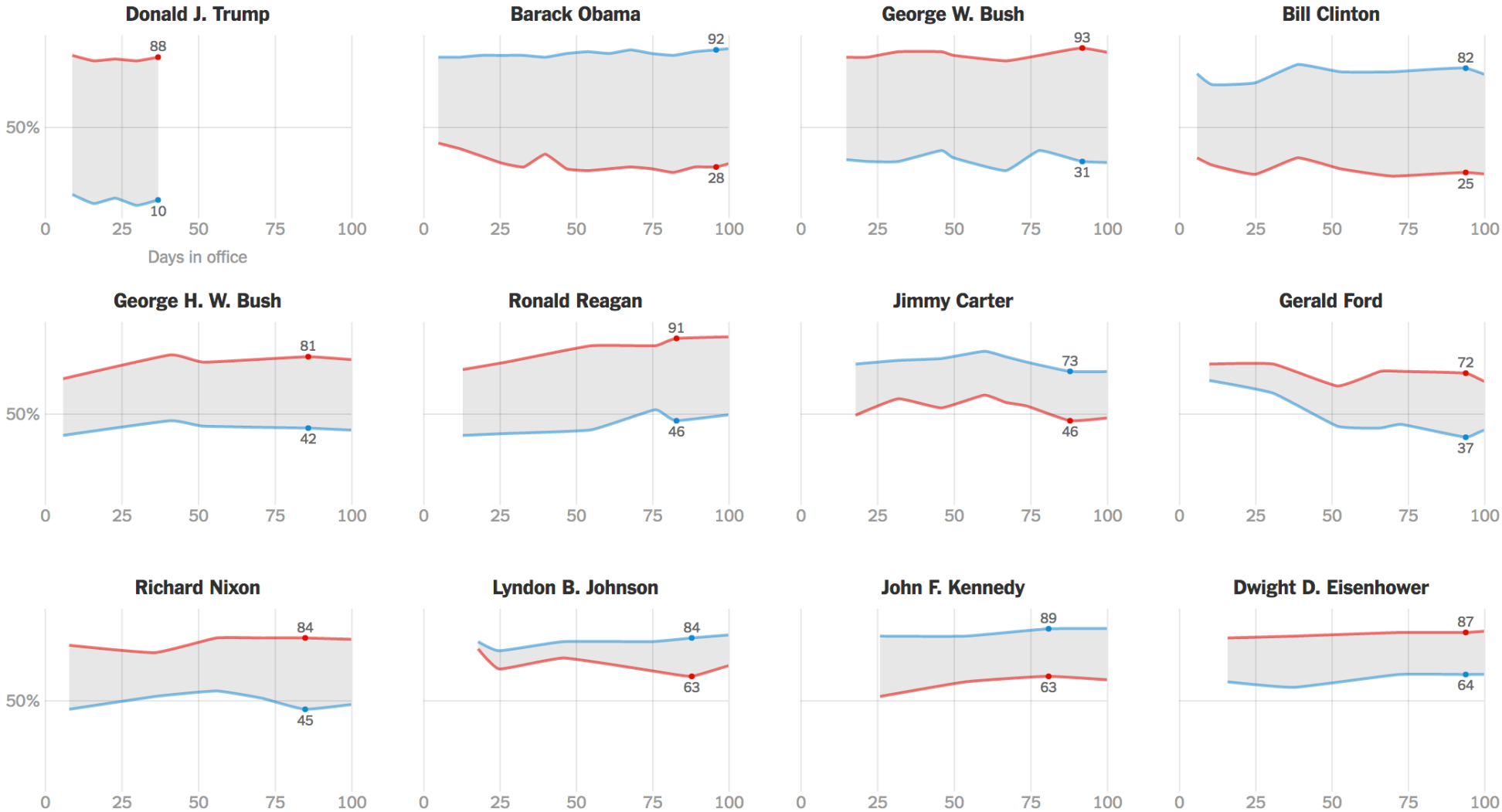




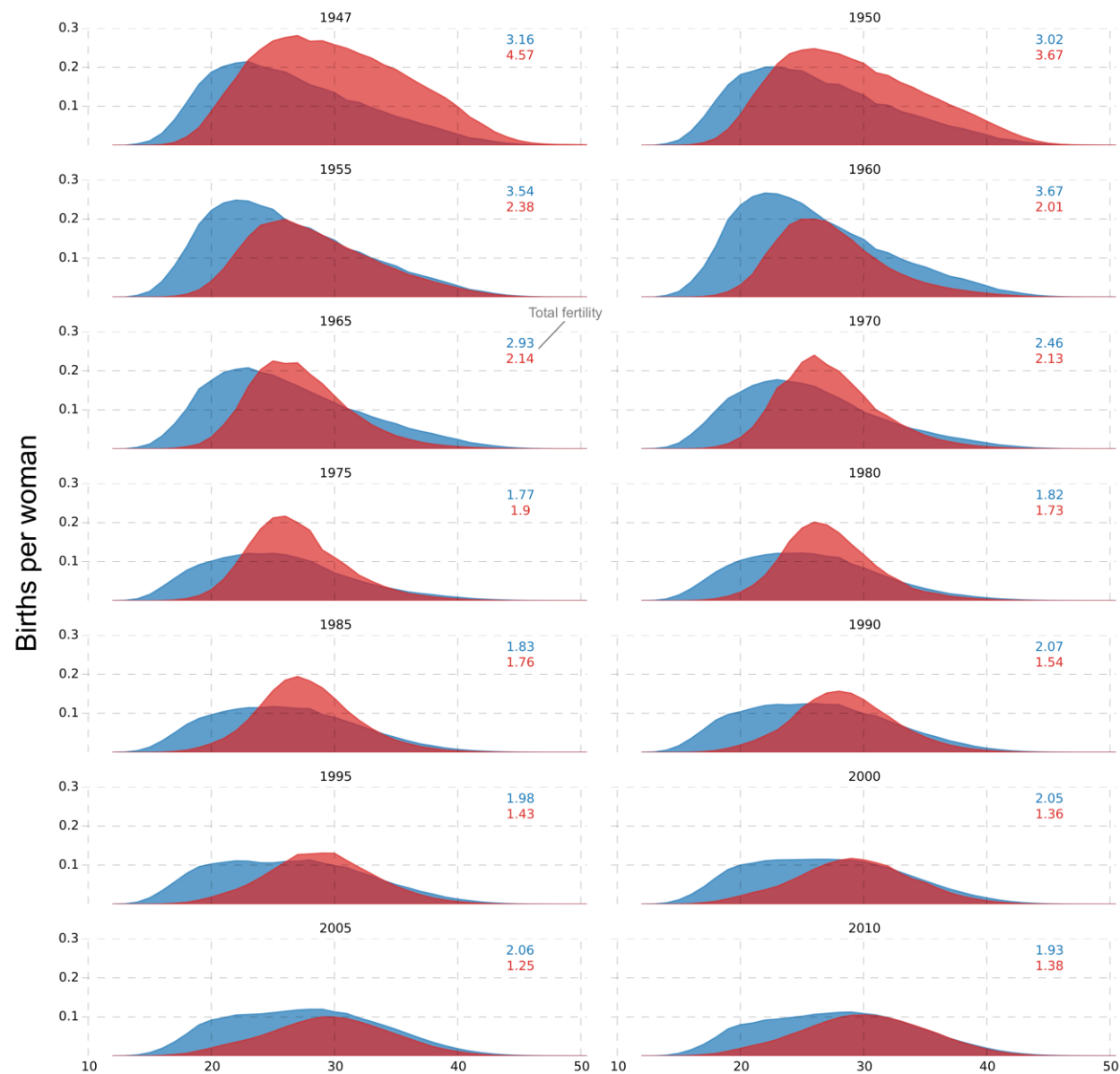




A Record Approval Gap Between Republicans and Democrats



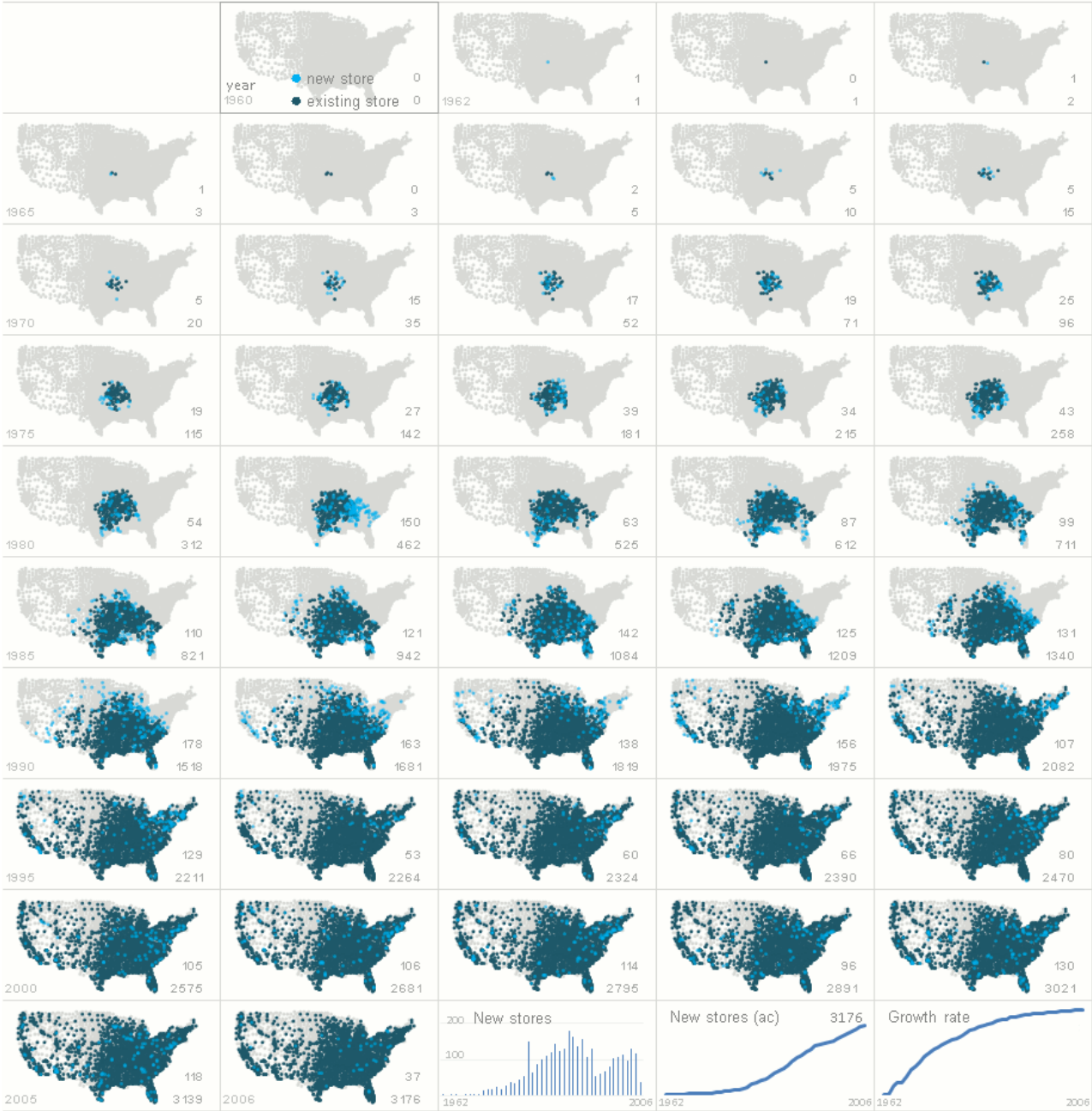
Fertility in USA and Japan, 1947 - 2010



Data source: Human Fertility Database (humanfertility.org)
 Author: Randy Olson (randalolson.com / @randal_olson)

Age when giving birth

Growth of Walmart



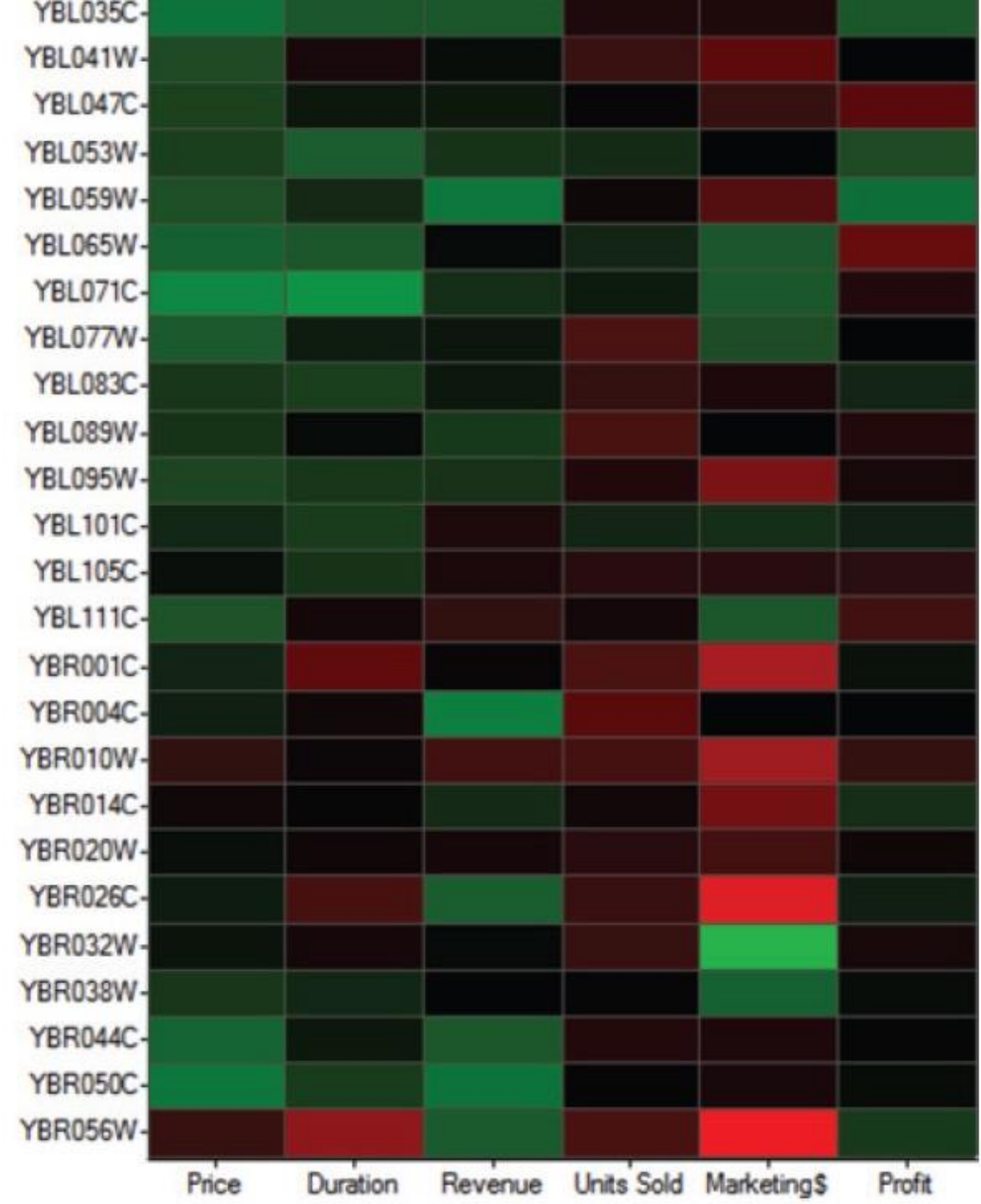
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Color-blindness!

Black used here to encode average values is **too visually prominent**.

Difficult to use when searching for **similar profiles**.

Can be used to **reveal exceptions** (bright green/red values in Marketing\$ column).

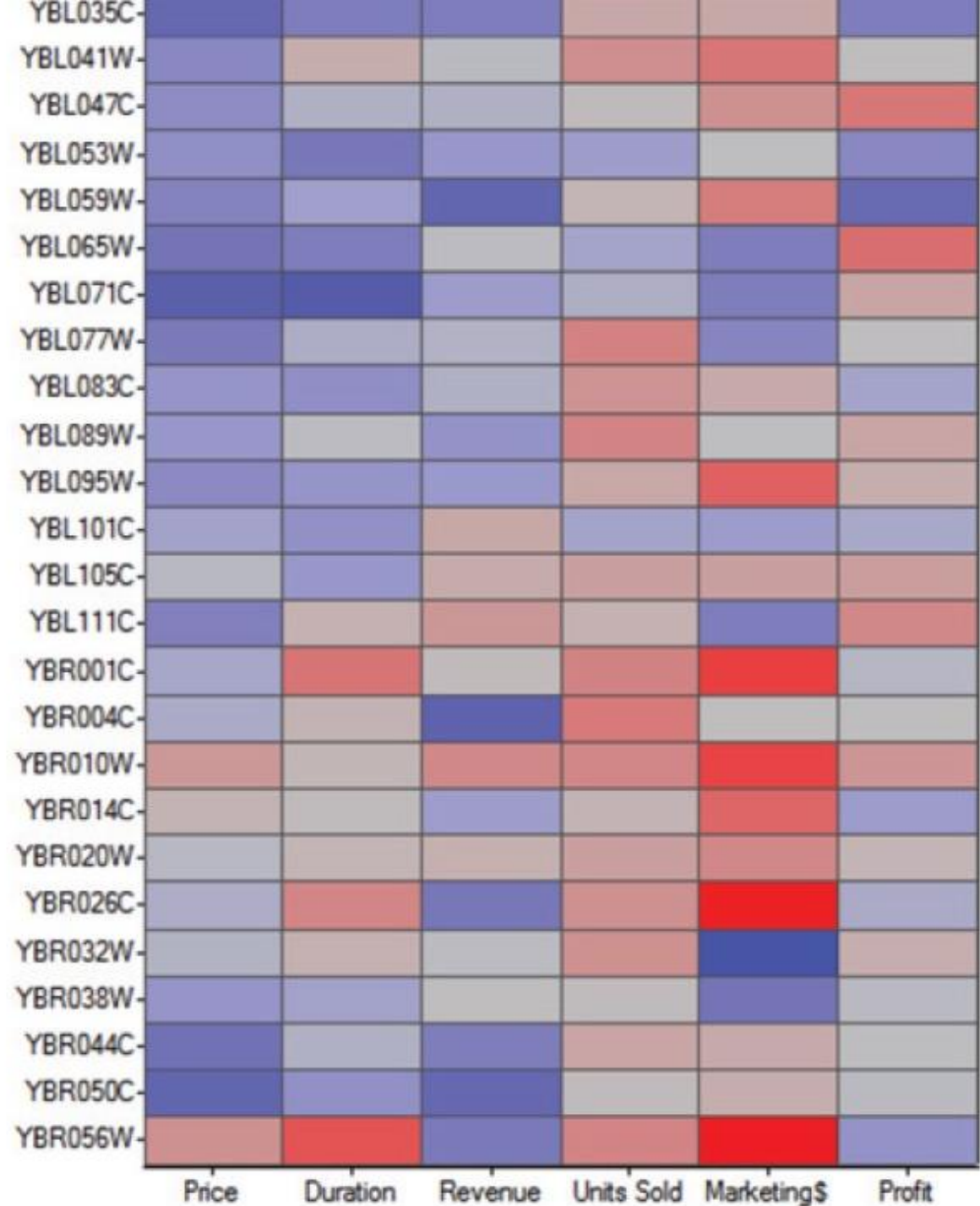


Positive values are **blue**,
negative values are **red**.

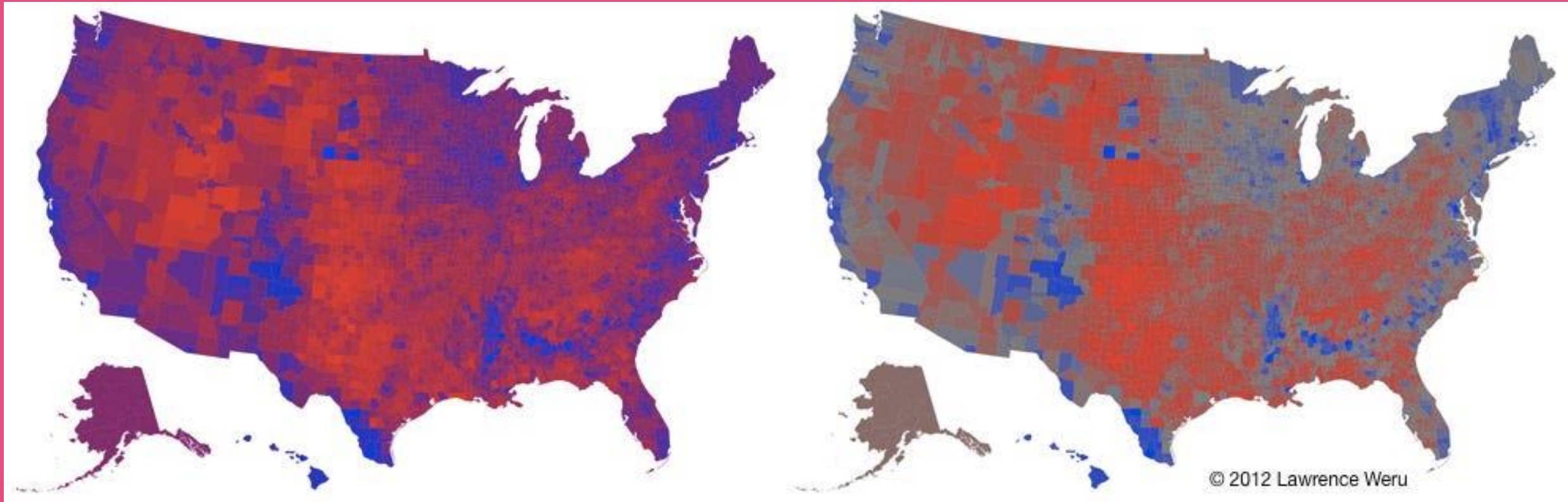
Average values fade to light
gray.

Vibrant reds and blues **grab our
attention** (and represent
extreme values).

Still can't see patterns in data!



Remember this?

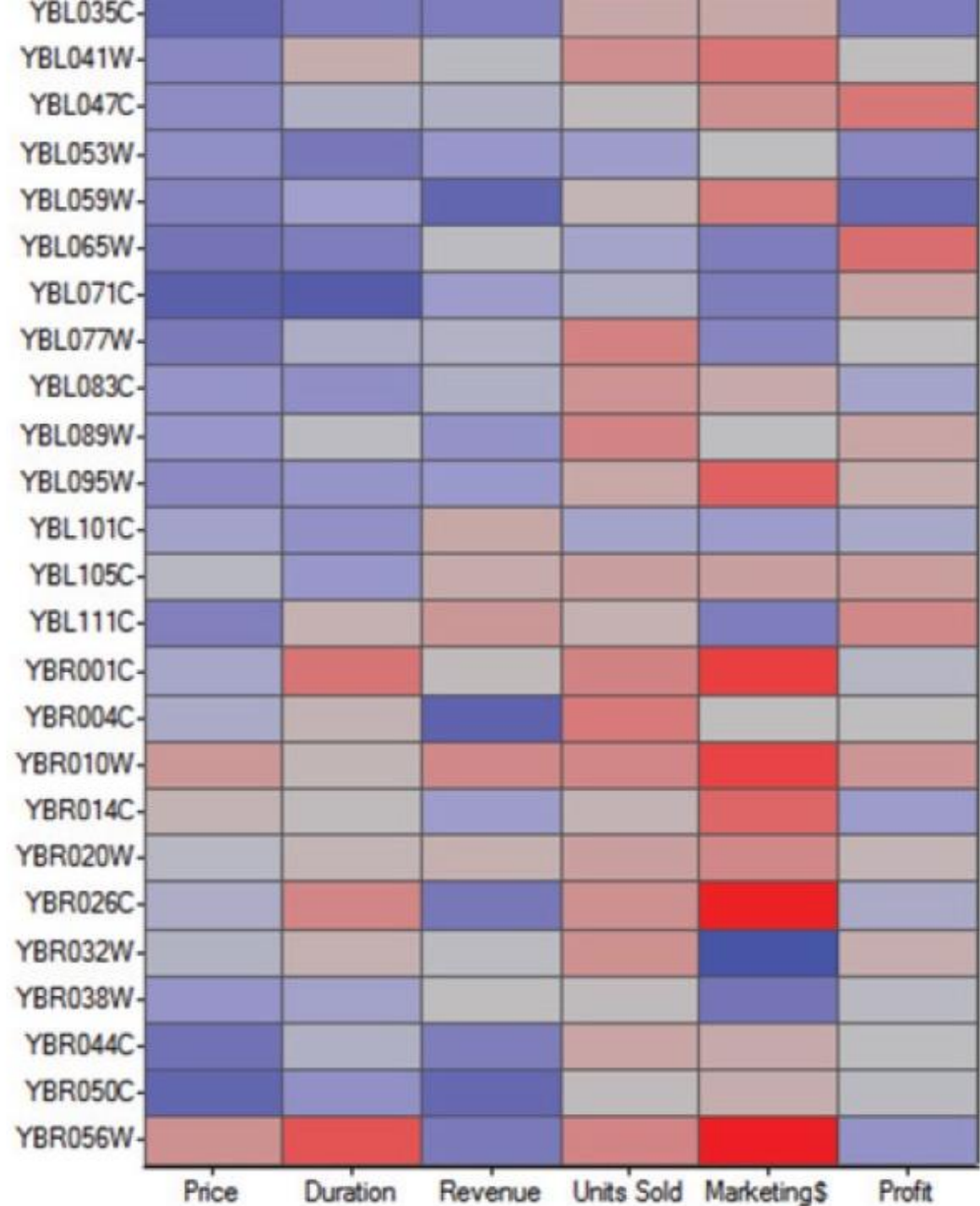


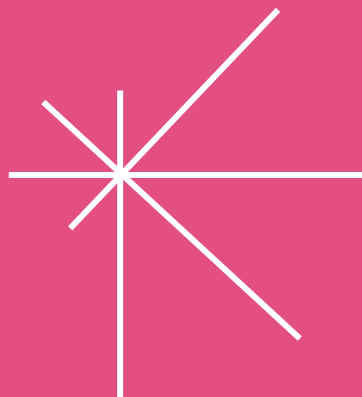
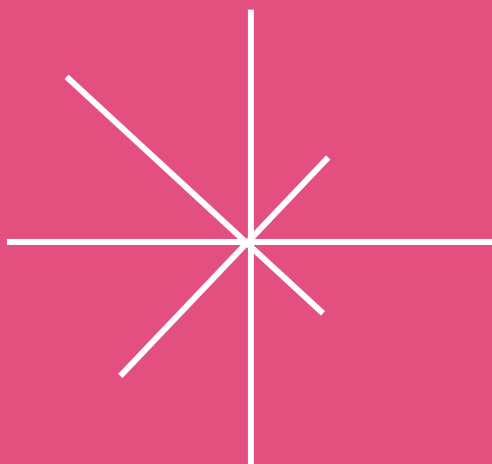
Positive values are **blue**,
negative values are **red**.

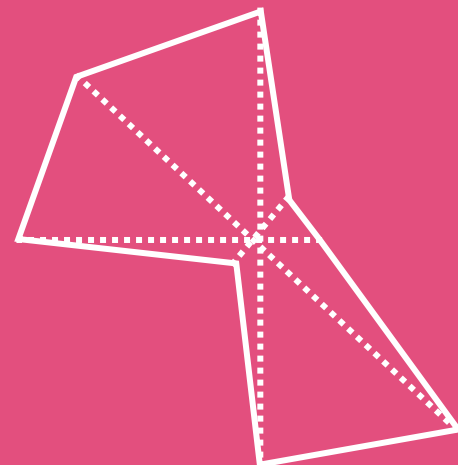
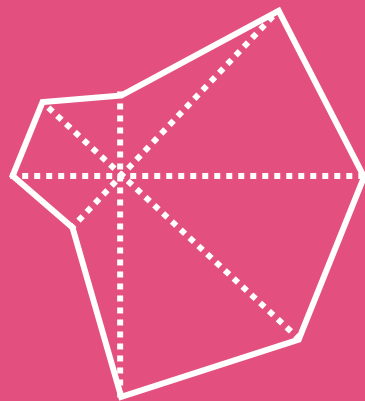
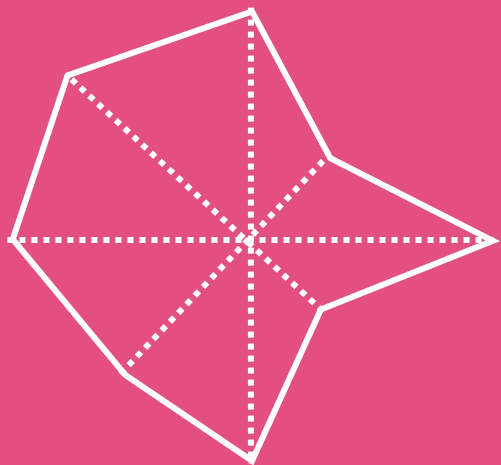
Average values fade to light
gray.

Vibrant reds and blues **grab our
attention** (and represent
extreme values).

Still can't see patterns in data!

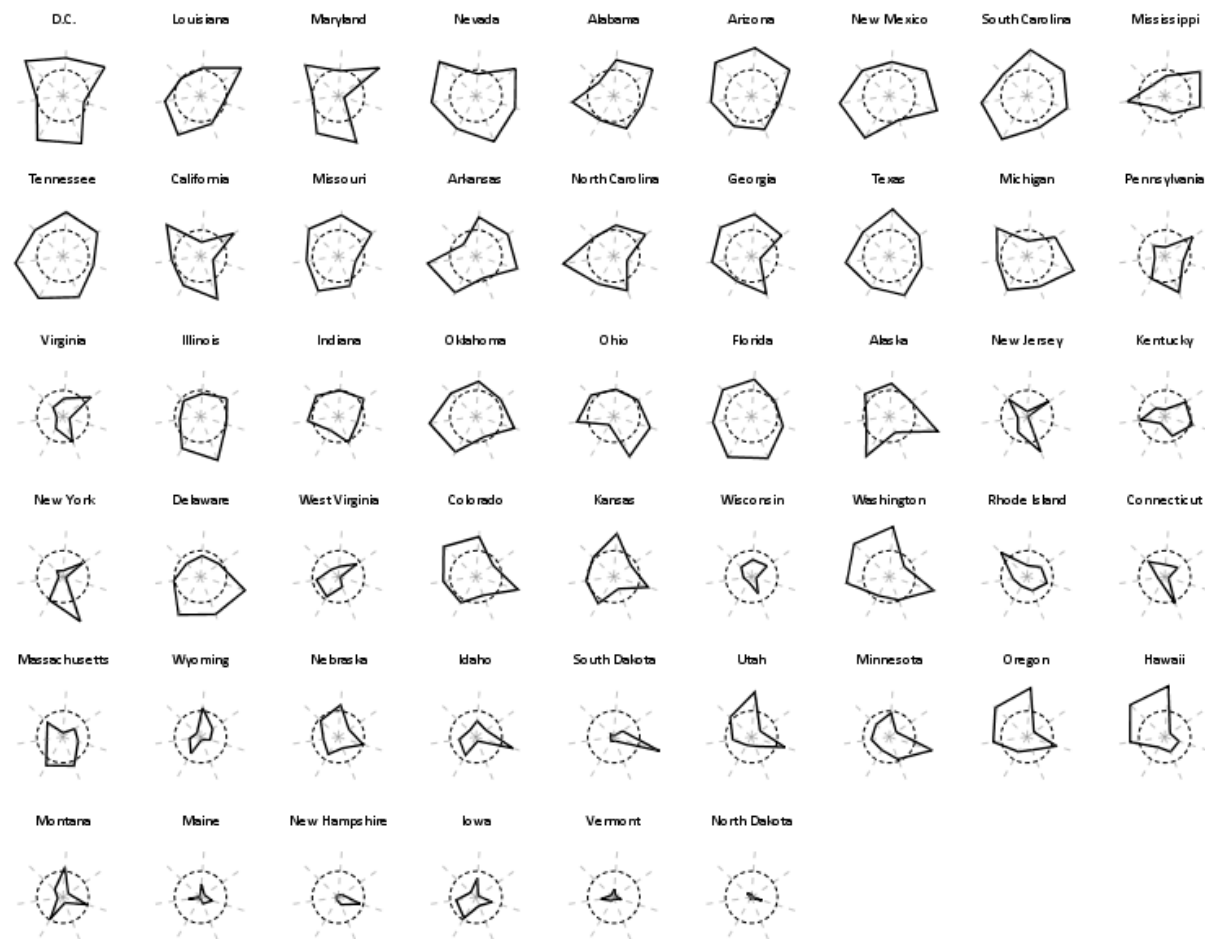






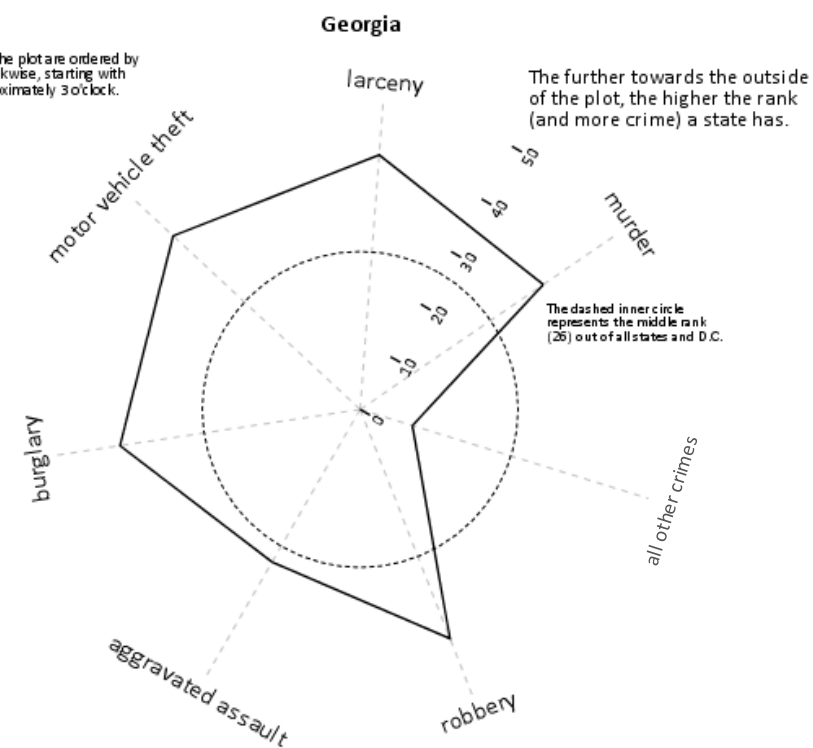
Rankings of Crime Rates for 50 States (and D.C.) Star Plot

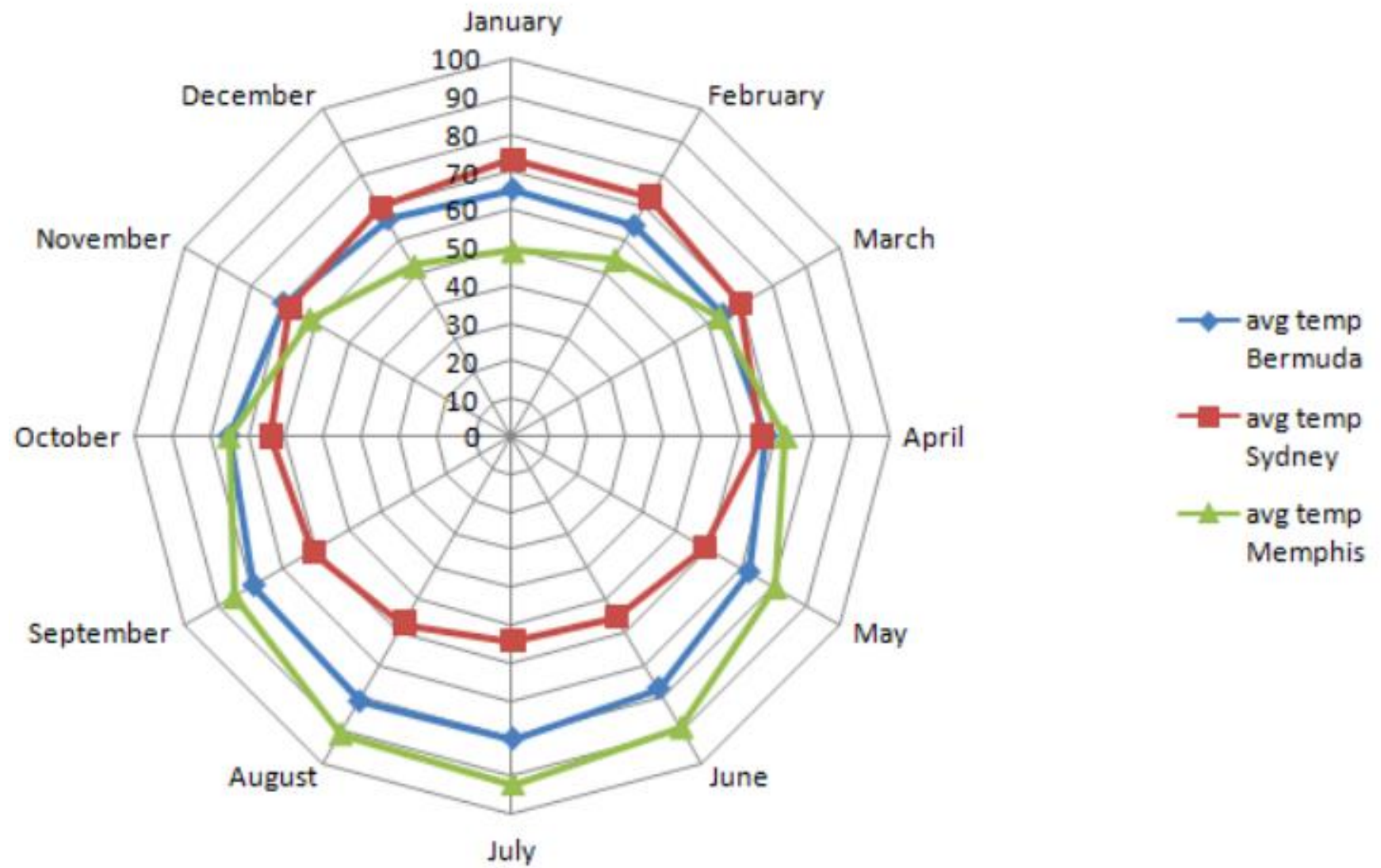
States ordered by homicide ranking in plot (left to right)

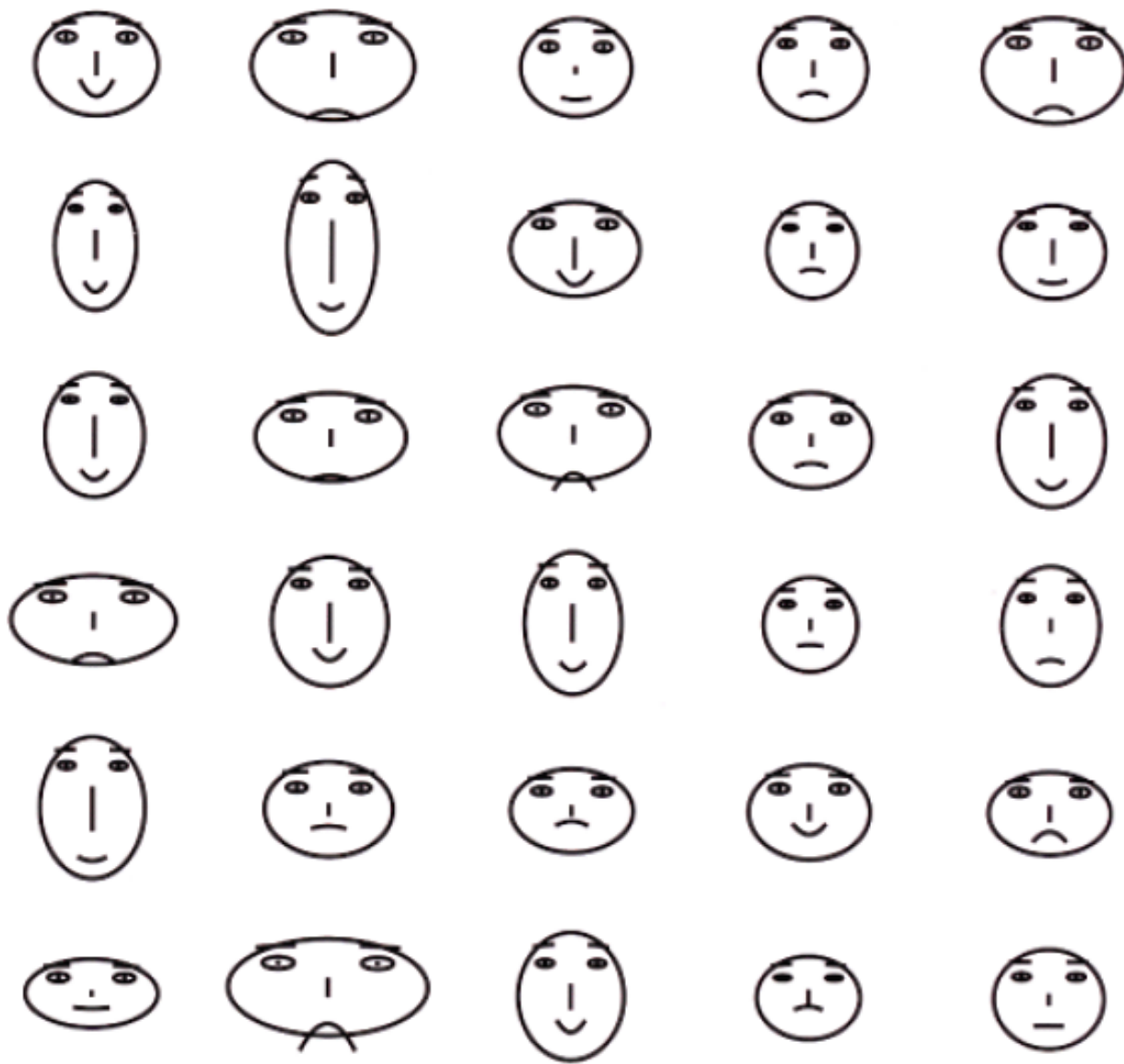


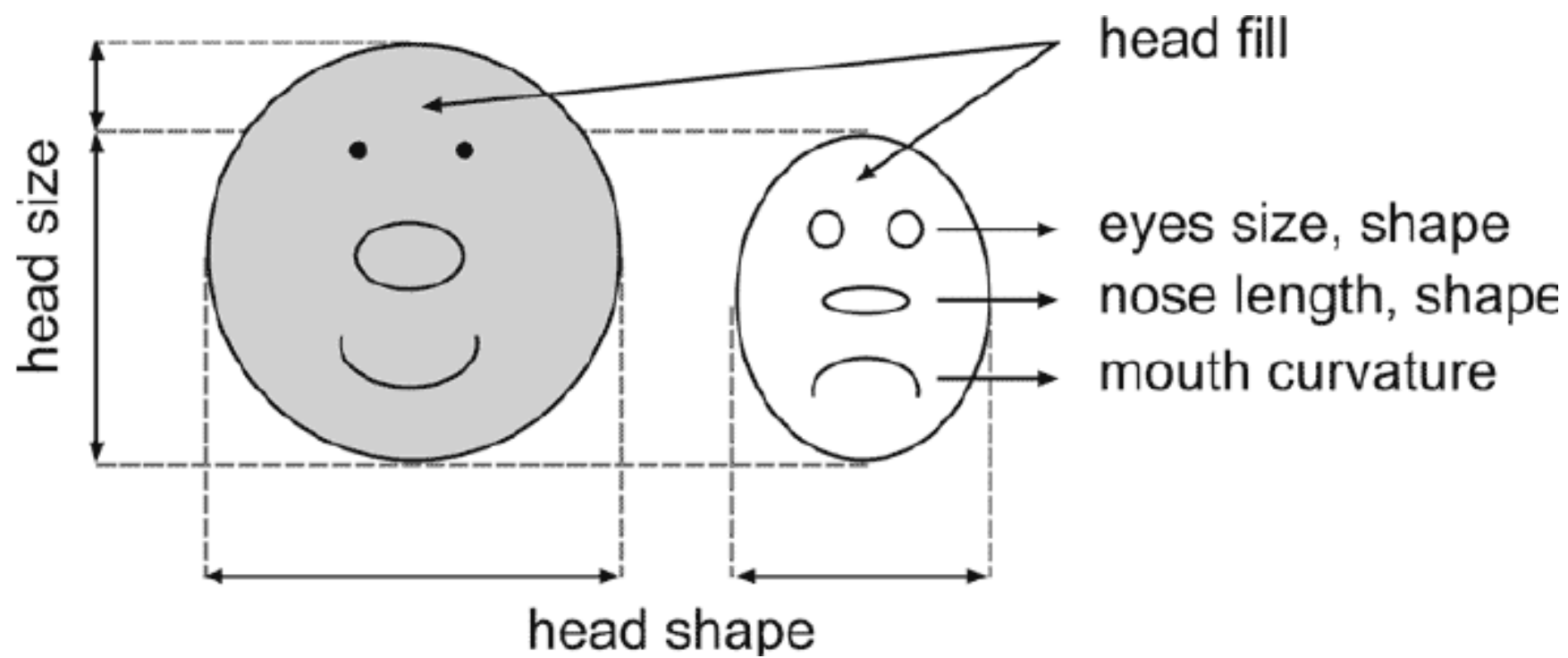
Annotated Legend for Star Plots

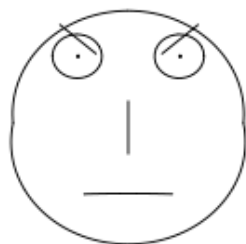
Crimes around the plot are ordered by seriousness clockwise, starting with murder at approximately 3 o'clock.



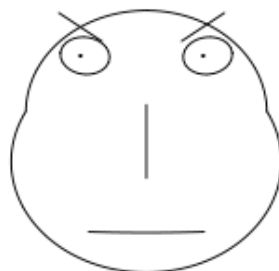




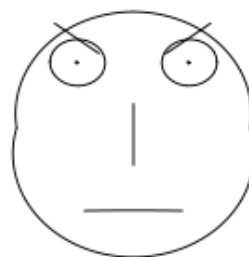




AARONSON, L.H.



ALEXANDER, J.M.



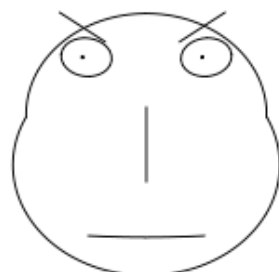
ARMENTANO, A.J.



BERDON, R.I.



BRACKEN, J.J.



BURNS, E.B.



CALLAHAN, R.J.



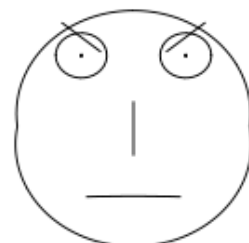
COHEN, S.S.



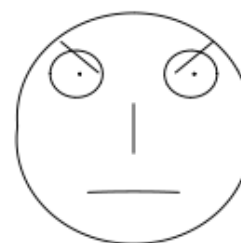
DALY, J.J.



DANNEHY, J.F.

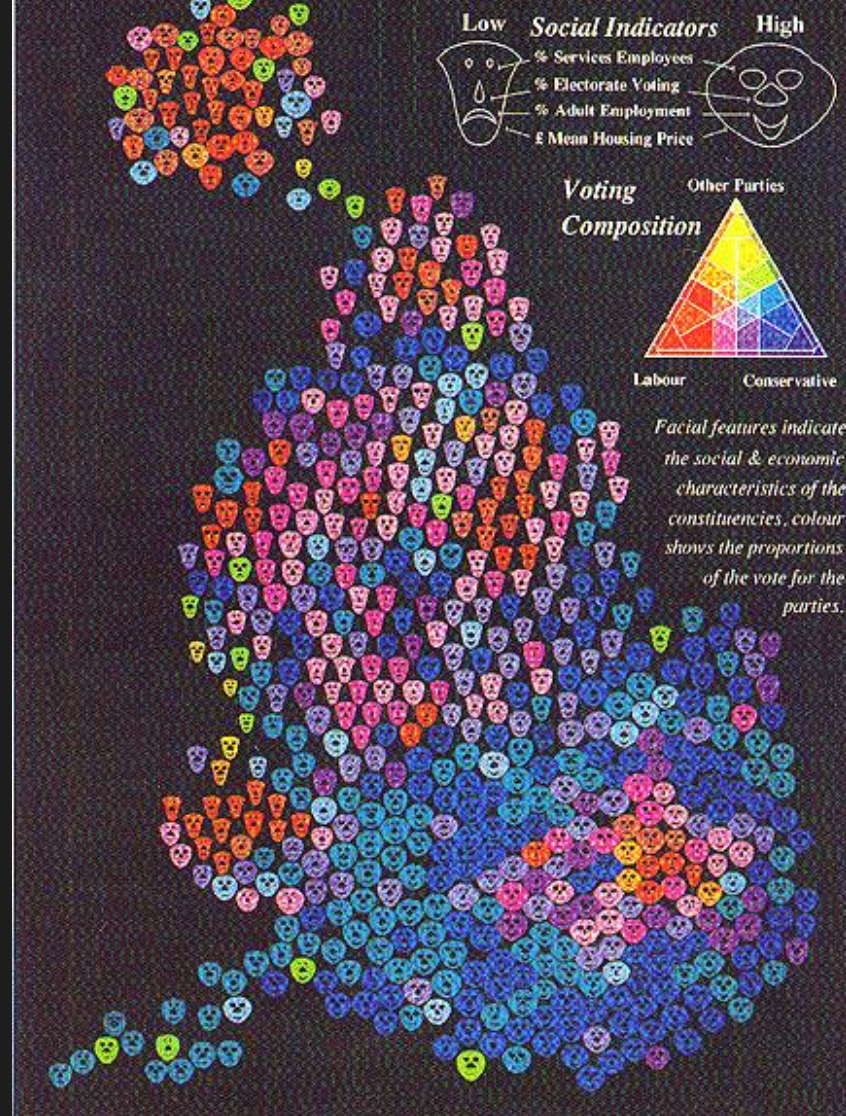


DEAN, H.H.

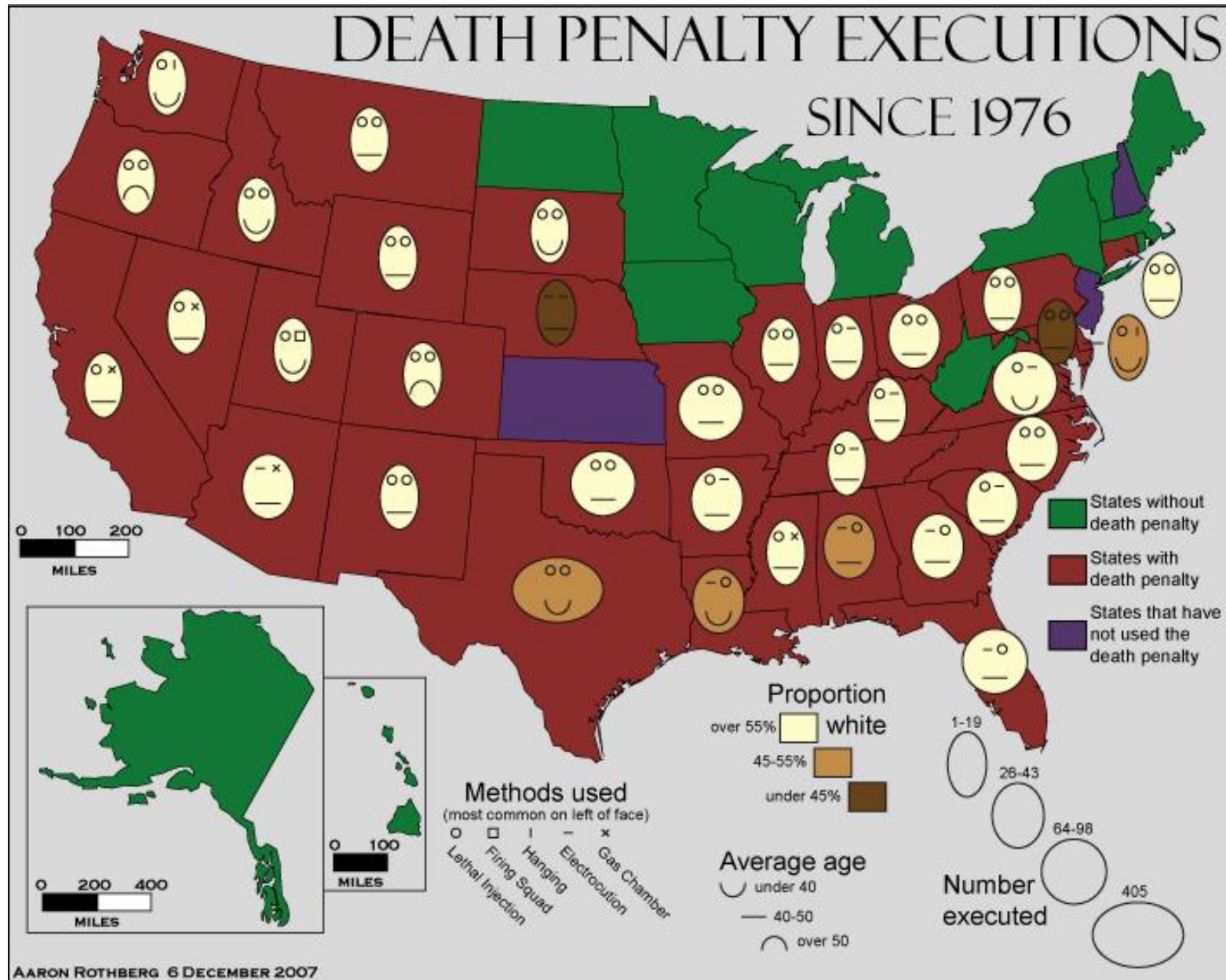


DEVITA, H.J.

The Distribution of Voting, Housing, Employment and Industrial Compositions in the 1983 General Election.



DEATH PENALTY EXECUTIONS SINCE 1976

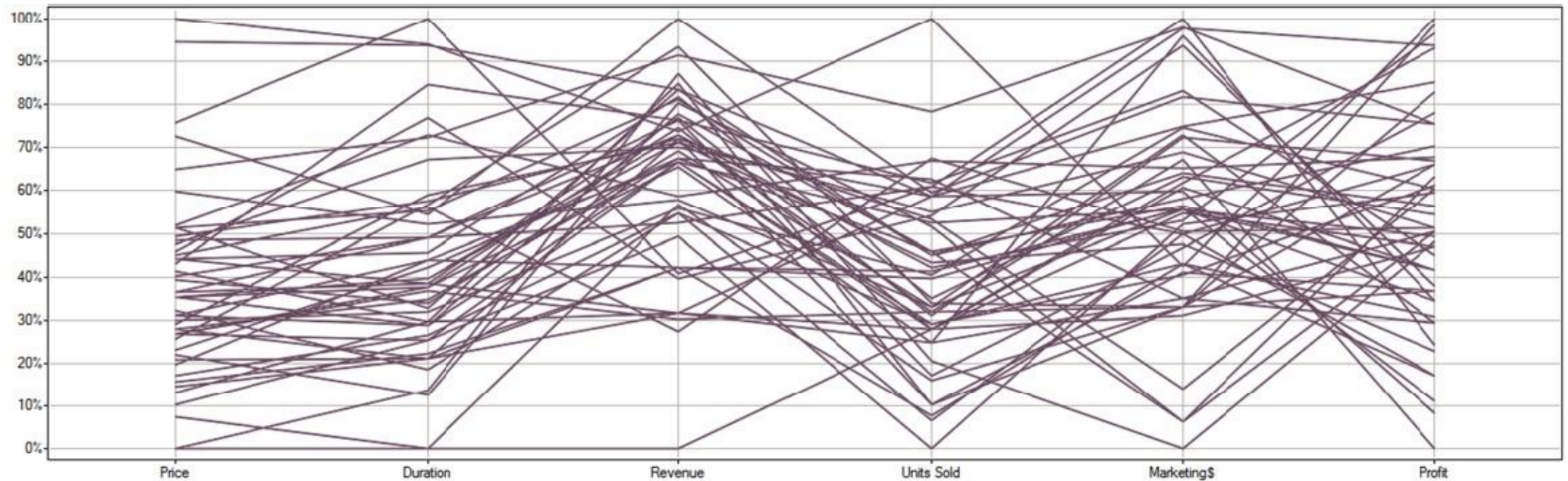


In a controlled study*, Chernoff faces and star glyphs led to slow, inaccurate answers being given with low confidence.

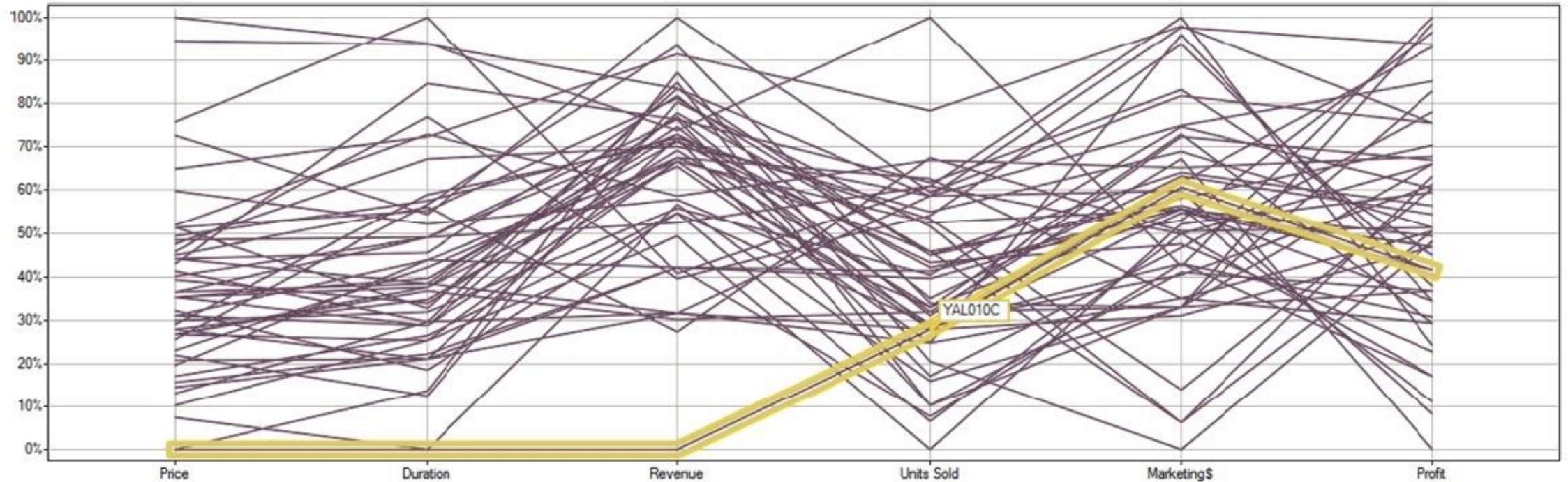
Spatial representations (where objects are points in multi-dimensional space) led to faster and more confident answers.

* Michael Lee, Rachel Reilly, Marcus Butavicius, 2003, "An Empirical Evaluation of Chernoff Faces, Star Glyphs, and Spatial Visualizations for Binary Data," Proc. Asia-Pacific Symposium on Information Visualization

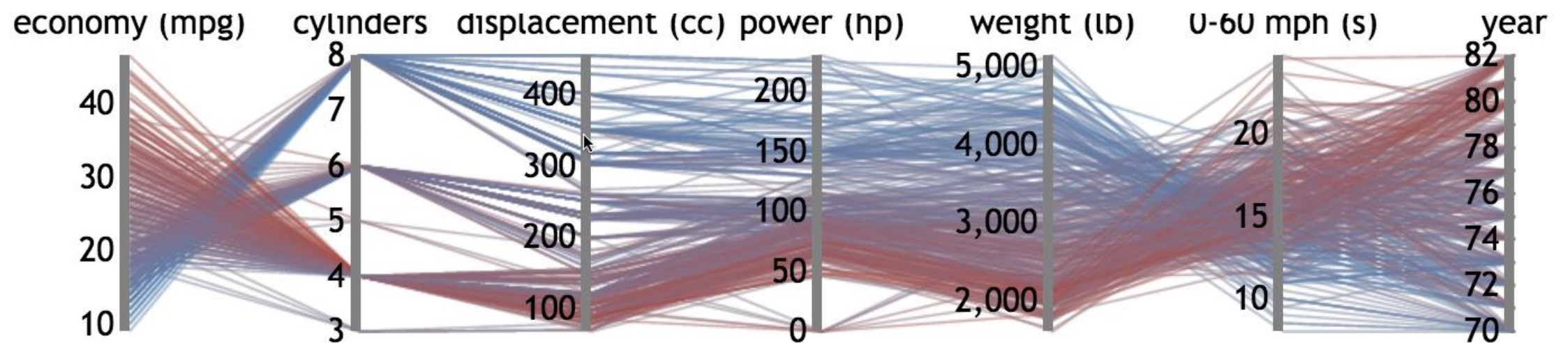
Parallel Coordinates



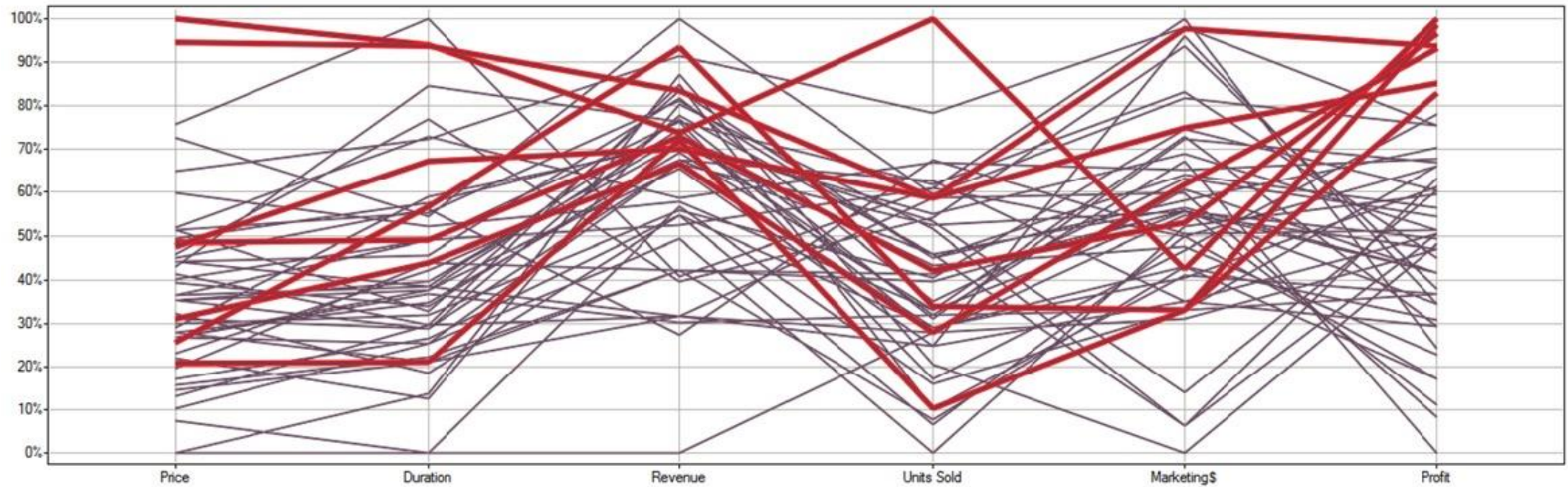
Meant for interactive exploration to identify patterns and outliers.



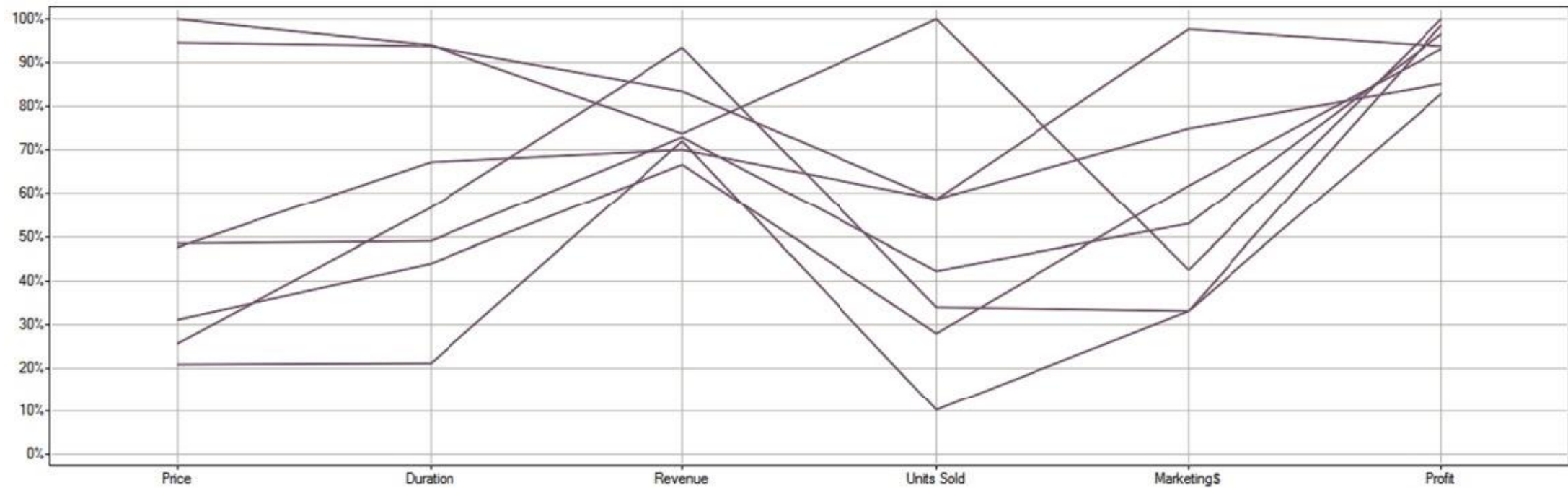
Directing attention to **subsets** within the data.



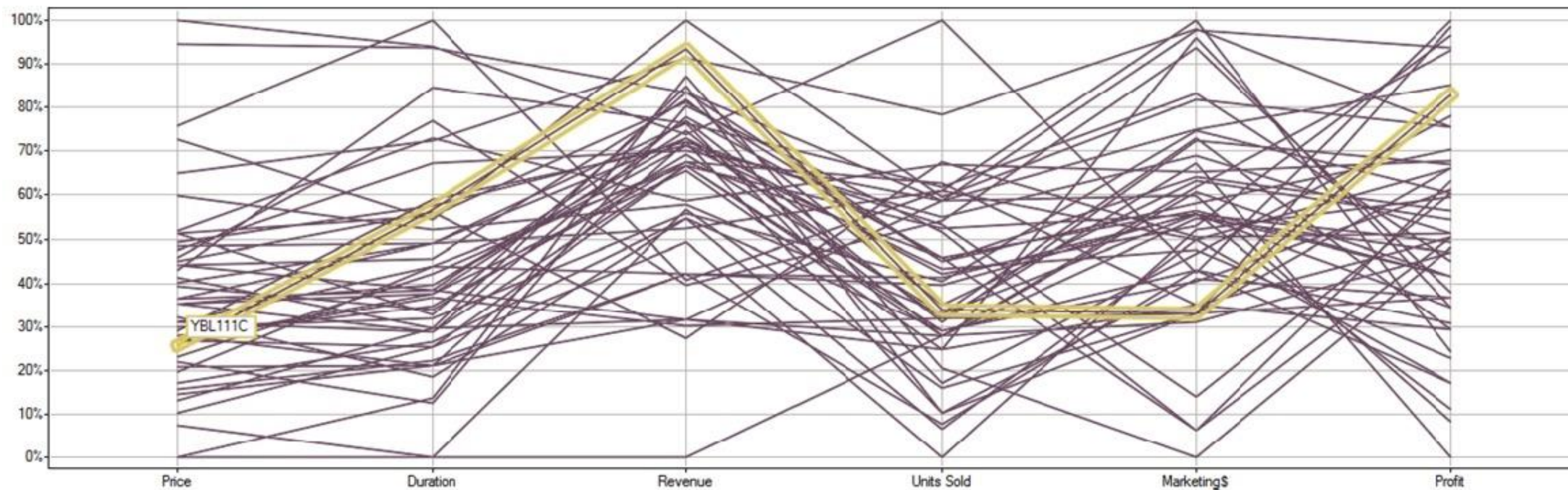
Highlighting/brushing all products with profits above 80%.



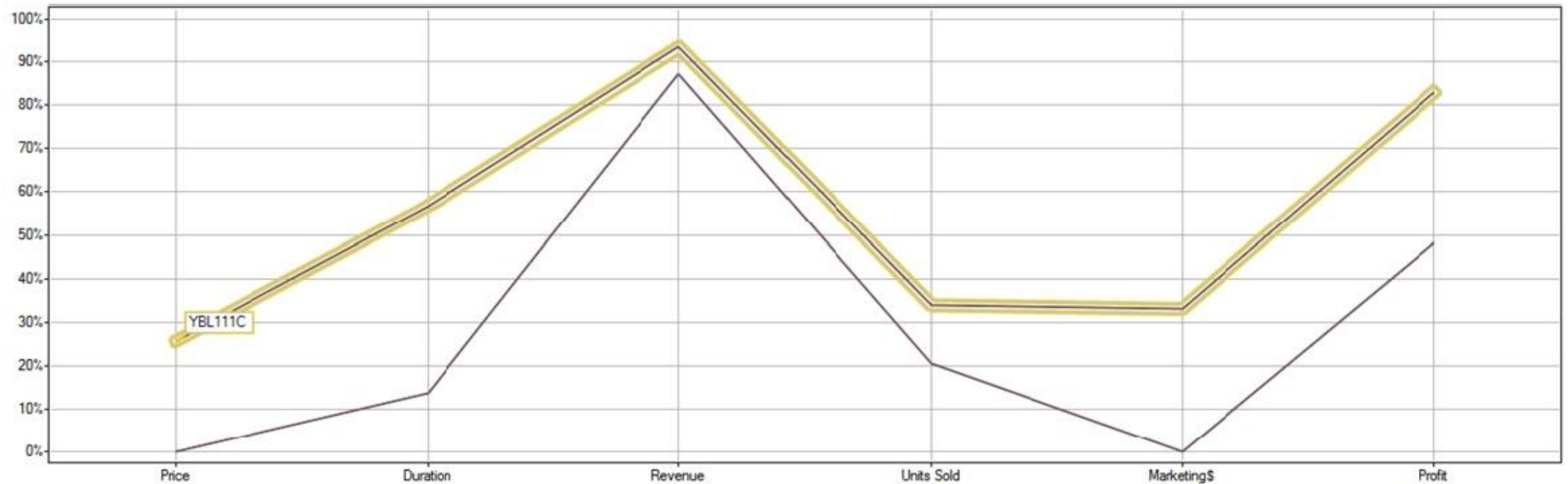
Filtering to bring the data of interest into focus.



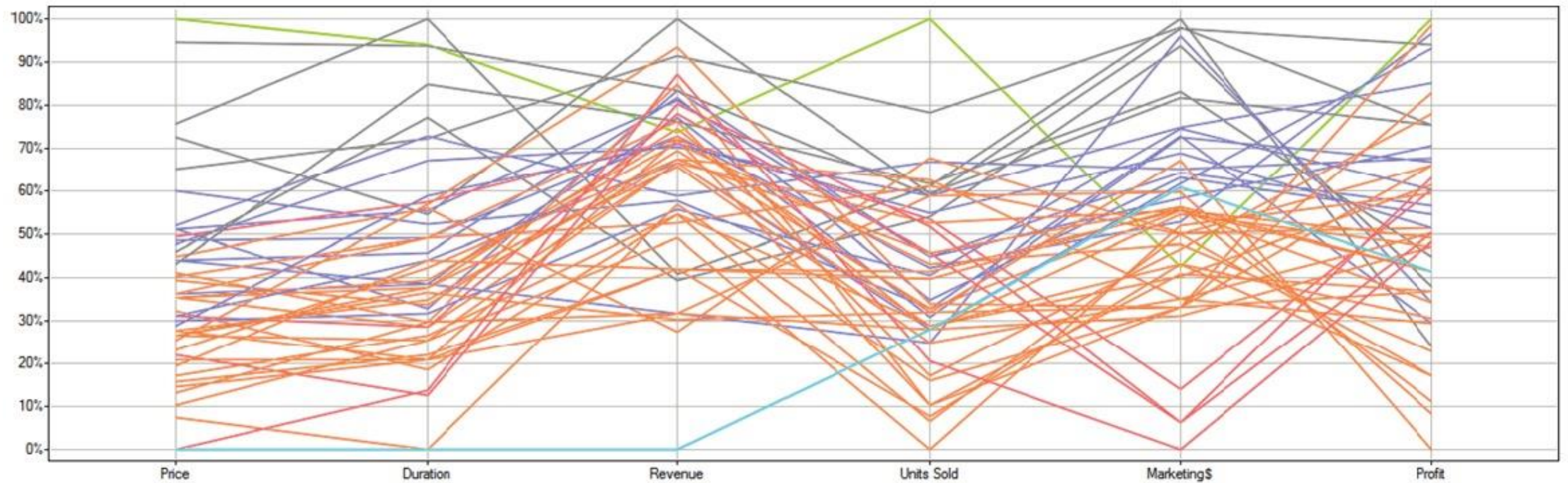
To find a particular pattern of data – ranking by similarity – we can find the best match to that data.

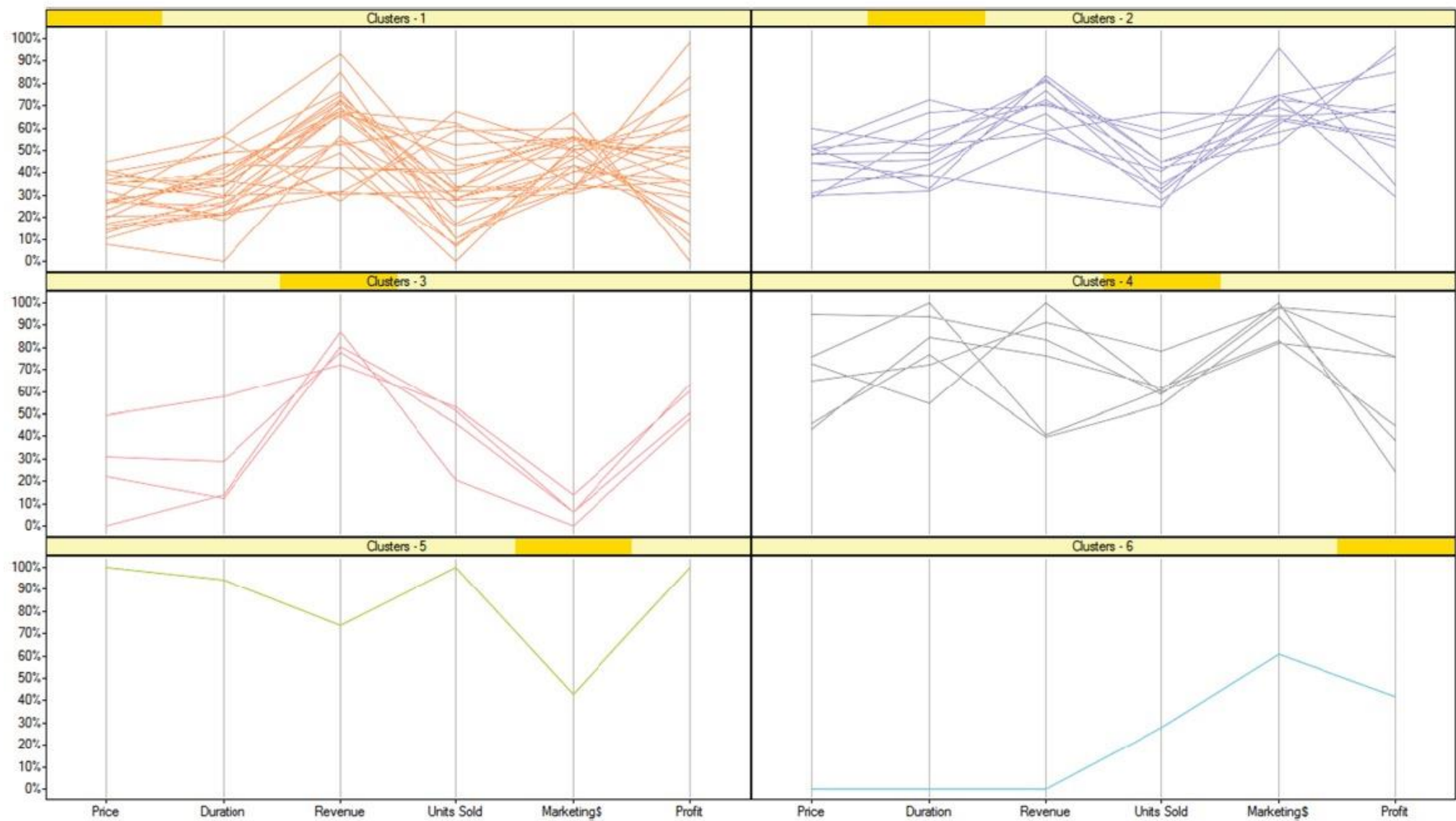


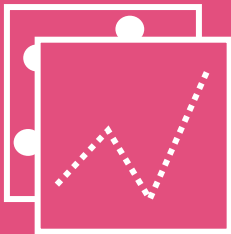
Filtering to show the **most similar** results.



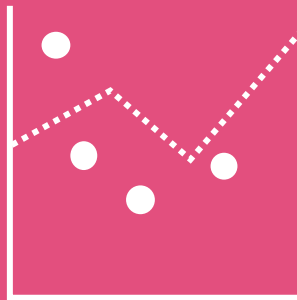
Clustering to segment data.



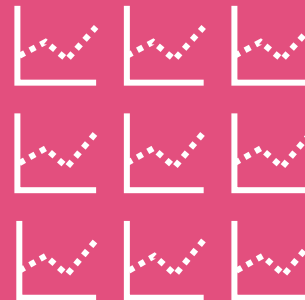




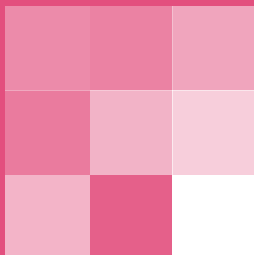
View Switching



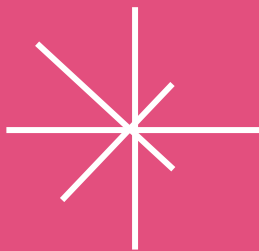
Layering



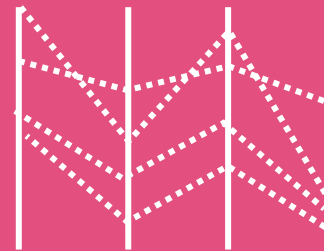
Small Multiples



Heatmaps



Glyphs



Parallel Coordinates

Break

Break for 20min

Activity

In your project groups:

Visualize Cars dataset as multivariate

You can use any tool to do this: Tableau, Excel, D3, etc

Questions:

What story do you want to tell?

What insight do you want to provide to users?

What data could be combined to tell this story?

How might you visualize this?

Reference the canvas activity

<https://canvas.uw.edu/courses/1862679/assignments/11046841>

Activity

Model	Origin	Year	Cylinders	Horsepower	MPG	Weight
amc ambassador dpl	US	70	8	190	15	3,850
chevrolet vega (sw)	US	71	4	72	22	2,408
chevrolet vega	US	72	4	90	20	2,408
ford pinto runabout	US	72	4	86	21	2,226
mazda rx2 coupe	Japan	72	3	97	19	2,330
amc matador (sw)	US	72	8	150	15	3,892
ford pinto (sw)	US	72	4	86	22	2,395
dodge coronet custom	US	73	8	150	15	3,777
toyota carina	Japan	73	4	88	20	2,279
chevrolet vega	US	73	4	72	21	2,401
datsum 610	Japan	73	4	94	22	2,379
ford pinto	US	73	4	85	19	2,310
chevrolet chevelle malibu classic	US	74	6	100	16	3,781
buick century	US	75	6	110	17	3,907
chevroelt chevelle malibu	US	75	6	105	16	3,897
plymouth fury	US	75	6	95	18	3,785
amc matador	US	76	8	120	15.5	3,962
mercedes-benz 280s	Europe	76	6	120	16.5	3,820
chevrolet caprice classic	US	77	8	145	17.5	3,880
chevrolet caprice classic	US	79	8	130	17	3,840
mercury grand marquis	US	79	8	138	16.5	3,955
dodge st. regis	US	79	8	135	18.2	3,830
vw rabbit	Europe	80	4	76	41.5	2,144
mazda glc	Japan	80	4	65	46.6	2,110
vw pickup	Europe	82	4	52	44	2,130

Activity

For 35-40 min, in your project groups:

Discuss which multivariate visualizations you may utilize to address your users' goals

**Sketch them out:
whiteboard/paper/digital**

Then prepare to demo / share out with the class for 3-5min

Group Time

**Time permitting tonight
work in your groups
on final project**